Terms Rewrite Systems Equational Reasoning, Complete Property

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Abstract. Term rewriting systems (TRSs) provide a computational model for equational reasoning, where equations are oriented into directed rules and applied as local transformations. A central question in this setting is whether a TRS is complete with respect to an underlying equational theory, that is, whether rewriting alone is sufficient to decide equational validity. This paper surveys the main notions and methods surrounding this property. We first recall the standard framework of TRSs and equational theories, then examine the conditions under which completeness holds, with particular attention to how it can be preserved under disjoint unions of systems. We review classical results connecting completeness to confluence and termination, discuss criteria for detecting its failure, and present the principal techniques for restoring it, including syntactic unification and Knuth–Bendix completion. This survey aims to give a unified and accessible account of completeness in term rewriting, serving as a reference for readers interested in the foundations of equational logic and automated deduction.

Keywords. term rewriting, confluence, termination, normalisation, complete, unification, modularity, Knuth–Bendix completion, equational reasoning.

Contents

1 Introduction

Equational reasoning constitutes one of the foundational pillars of mathematical logic and theoretical computer science, providing a uniform framework for describing the algebraic properties of computational structures. Term Rewriting Systems (TRS) offer a particularly elegant and powerful approach to this problem, by orienting equational axioms into directed rewrite rules, thereby transforming the undirected question of equational provability into a deterministic computation of normal forms. Central to this framework is the notion of confluence: the property ensuring that any two reduction sequences issuing from a common term eventually converge to a single result. Far from being a mere technical convenience, confluence is the cornerstone upon which the entire decision-theoretic apparatus of equational reasoning rests: it guarantees the uniqueness of normal forms, ensures the coherence of the induced equational theory, and, when combined with termination, yields the decidability of the word problem.

Our guiding goal throughout this paper is *equational reasoning itself*: given a set of equational axioms, we want an effective, mechanical procedure to decide whether two terms are provably equal. Term rewriting is the vehicle for this goal, and confluence together with termination is exactly what makes the vehicle run: it is what turns the static congruence $=_E$ into a computable equivalence, decidable by comparing normal forms (Section 3). Everything else in this paper is in the service of that single idea. We first develop the classical machinery for establishing confluence: Newman's Lemma, syntactic unification, and the Critical Pair Theorem (Sections 3 to 5), and the constructive procedure that restores confluence when it fails, Knuth–Bendix completion (Section 6). We then turn to a more peripheral, though instructive, question: how these properties behave under disjoint unions of systems (Section 7), where confluence turns out to be modular but termination is not. We close with a fundamental limit on the whole programme: termination of an arbitrary TRS is itself undecidable (Section 8).

Reading this paper. Section 2 lays down the syntactic apparatus (terms, substitutions, equational theories, positions, rewrite steps). Section 3 develops confluence, termination, and their interaction via Newman's Lemma, the technical heart of the equational-reasoning programme. Section 4 develops syntactic unification, a prerequisite for the algorithmic construction of critical pairs, presented immediately afterwards in Section 5 together with the Critical Pair Theorem. Section 6 presents the Knuth–Bendix completion procedure, which effectively restores confluence using critical pairs, and its correctness theorem: at this point, the central goal of the paper (an effective decision procedure for equational validity) is fully achieved. Section 7 is a secondary study of how confluence, termination, and convergence behave under disjoint unions of systems; it is not required for the main line of the paper and may be skipped on a first reading. Section 8 is a self-contained appendix-like section showing, via an encoding of Turing machines, that termination of an arbitrary TRS is undecidable.

2 Preliminaries

2.1 Terms and signatures

> **Definition** (*signature*). A *signature* Σ is a set of function symbols, each equipped with an *arity* $\mathrm{ar} : \Sigma \to \mathbb{N}$. Symbols of arity 0 are called *constants*.

Nota. We fix two denumerable sets $\mathcal{X}$ and $\mathcal{C}$, of *variables* and *constants* respectively, disjoint from Σ and from each other.

Definition (*Terms*). A *term* over Σ and a set of atoms $\mathcal{A}$ (either $\mathcal{X}$ or $\mathcal{C}$) is defined inductively: every element $a \in \mathcal{A}$ is a term, and if $f \in \Sigma$ with $\mathrm{ar}(f) = n$ and $t_1, \ldots, t_n$ are terms, then $f(t_1, \ldots, t_n)$ is a term.

Note. We write $\mathcal{T}(\Sigma, \mathcal{X})$ for the set of *patterns* (terms built over the variables $\mathcal{X}$), and $\mathcal{T}(\Sigma, \mathcal{C})$ for the set of *ground terms* (terms built over the constants $\mathcal{C}$, containing no variable).

Example. Let $\Sigma = \{0, s, +\}$ with arities $0, 1, 2$ respectively (0 a constant, s a unary successor symbol, $+$ a binary addition symbol). Let $\mathcal{C} = \{a, b, c, \ldots\}$ and $\mathcal{X} = \{x, y, z, \ldots\}$.

- *Ground terms* ($\mathcal{T}(\Sigma, \mathcal{C})$, no variable): 0, $s(0)$, $s(s(0))$, $+(s(0), s(s(0)))$, and also $+(a, s(b))$ once $a, b \in \mathcal{C}$ are used as free constants.
- *Patterns* ($\mathcal{T}(\Sigma, \mathcal{X})$, possibly containing variables): $+(x, s(0))$, $s(s(y))$, $+(x, +(y, z))$.

Ground terms are the terms with no variable occurrence; patterns are the general syntactic objects manipulated by rewrite rules, whose variables get instantiated by substitution.

Definition (*Substitution*). A *substitution* σ is a map $\sigma : \mathcal{X} \to \mathcal{T}(\Sigma, \mathcal{X})$.

Note. Generally, we want σ to have finite support, i.e.

$$|\{x \in \mathcal{X} : \sigma(x) \neq x\}| < +\infty.$$

Note (extension). A substitution does not apply only to variables, but to all terms, by homomorphism: it extends uniquely to a map compatible with every function symbol of Σ, as follows (this is exactly the notion of compatibility with operations that Definition [Congruence], below, makes precise in general):

- if x is a variable: $x\sigma = \sigma(x)$;
- if $f(t_1, \ldots, t_n)$ is a term: $f(t_1, \ldots, t_n)\sigma = f(t_1\sigma, \ldots, t_n\sigma)$.

2.2 Equational theories

Definition (*Congruence*). Let A be a set equipped with a family of operations $(\omega_i)_{i \in I}$, each $\omega_i : A^{n_i} \to A$ of arity n_i (this is called an *algebra*, or Ω-*algebra*, in the sense of universal algebra; groups, rings, and term algebras are all instances of this notion, each with their own choice of operations ω_i: for a group, $\Omega = \{\cdot, {}^{-1}, e\}$; for a term algebra over Σ, $\Omega = \Sigma$ itself, each symbol $f \in \Sigma$ acting as an operation of arity $\mathrm{ar}(f)$). A *congruence* on A is an equivalence relation $\sim$ on A that is *compatible* with every operation: for every $i \in I$ and every $a_1, \ldots, a_{n_i}, b_1, \ldots, b_{n_i} \in A$,

$$a_1 \sim b_1, \ \ldots, \ a_{n_i} \sim b_{n_i} \implies \omega_i(a_1, \ldots, a_{n_i}) \sim \omega_i(b_1, \ldots, b_{n_i}).$$

Definition (*Quotient set induced by a congruence*). Given a congruence $\sim$ on A, the *quotient set* $A/\!\sim$ is the set of equivalence classes $[a]_\sim = \{b \in A : a \sim b\}$ for $a \in A$. Each operation ω_i descends to a well-defined operation on $A/\!\sim$, by

$$\omega_i\big([a_1]_\sim, \ldots, [a_{n_i}]_\sim\big) := [\omega_i(a_1, \ldots, a_{n_i})]_\sim,$$

this definition being independent of the choice of representatives $a_1, \ldots, a_{n_i}$ precisely because $\sim$ is a congruence (this is exactly what compatibility guarantees).

Note. Specializing to $A = \mathcal{T}(\Sigma, \mathcal{X})$ with operations given by Σ itself (each $f \in \Sigma$ acting as an operation of arity $\mathrm{ar}(f)$), a congruence on terms is exactly an equivalence relation $\sim$ such that $t_i \sim v_i$ for every i implies $f(t_1, \ldots, t_n) \sim f(v_1, \ldots, v_n)$ for every $f \in \Sigma$. This is the notion of congruence used throughout this paper.

Definition (*Equational theory*). An *equational theory* E over a signature Σ is a set of equations $s \approx t$ with $s, t \in \mathcal{T}(\Sigma, \mathcal{X})$.

Definition (*Equational congruence induced by* E). Given an equational theory E, we write $=_E$ for the smallest congruence on $\mathcal{T}(\Sigma, \mathcal{X})$, stable under substitution, containing every instance $s\sigma \approx t\sigma$ for $s \approx t \in E$ and every substitution σ.

Note. Every equational theory E can be read as a term rewriting system, by orienting each equation $s \approx t$ into a directed rule pointing from one side to the other. This is precisely the purpose of term rewriting, introduced in the next section: to turn the undirected relation $=_E$ into a computable, directed process.

2.3 Rewrite rules

Definition (*Term Rewriting System (TRS)*). A *TRS* is a set R of rules $l \to_R r$ where $l, r \in \mathcal{T}(\Sigma, \mathcal{X})$, and we say that l *rewrites to* r. We also write (Σ, R) to say a set of rules on the alphabet or signature Σ. We require $\mathcal{V}\mathrm{ar}(r) \subseteq \mathcal{V}\mathrm{ar}(l)$ for every rule (the right-hand side introduces no variable absent from the left-hand side): this ensures that $r\sigma$ is fully determined by the restriction of σ to $\mathcal{V}\mathrm{ar}(l)$, i.e. by the match of l against the redex alone, so that a rewrite step has a uniquely determined result. Without this condition, a single redex could be rewritten to arbitrarily many different terms, depending on the (otherwise unconstrained) value of σ on the variables of r not occurring in l.

Nota (notation). For a set S, we write S^* for the set of all finite sequences (possibly empty) of elements of S, i.e. $S^* = \bigcup_{k \geq 0} S^k$ (the *Kleene star* of S); this includes the empty sequence ε. Below, we instantiate this with $S = \mathbb{N}_{>0} = \{1, 2, 3, \ldots\}$, the positive integers: $(\mathbb{N}_{>0})^*$ is thus the set of all finite sequences of positive integers, used to index the children of a node in a term (the i-th child, the j-th child of that child, and so on).

Definition (*Position and subterm at a position*). Let $t \in \mathcal{T}(\Sigma, \mathcal{X})$. We define simultaneously, by induction on the structure of t, the set $\mathrm{Pos}(t) \subseteq (\mathbb{N}_{>0})^*$ of *positions* of t (finite sequences of positive integers), and, for every $p \in \mathrm{Pos}(t)$, the *subterm of* t *at position* p, written $t|_p \in \mathcal{T}(\Sigma, \mathcal{X})$:

- If $t \in \mathcal{X} \cup \mathcal{C}$ (a variable or a constant):

$$\mathrm{Pos}(t) = \{\varepsilon\}, \qquad t|_\varepsilon = t.$$

- If $t = f(t_1, \ldots, t_n)$ with $n \geq 1$:

$$\mathrm{Pos}(t) = \{\varepsilon\} \cup \{\, i \cdot p \;:\; 1 \leq i \leq n,\ p \in \mathrm{Pos}(t_i) \,\}, \qquad t|_\varepsilon = t, \qquad t|_{i \cdot p} = t_i|_p.$$

The empty sequence ε always denotes the *root* of t, with $t|_\varepsilon = t$.

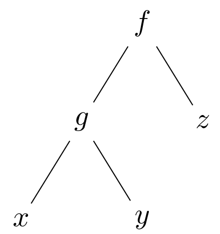
Example. Let $t = f(g(x,y),z)$. Then

$$\mathrm{Pos}(t) = \{\varepsilon,\ 1,\ 2,\ 1{\cdot}1,\ 1{\cdot}2\},$$

with

$$t|_\varepsilon = f(g(x,y),z), \quad t|_1 = g(x,y), \quad t|_2 = z, \quad t|_{1\cdot1} = x, \quad t|_{1\cdot2} = y.$$

Position 1 designates the first argument of f (here $g(x,y)$); positions starting with 1 then continue into the arguments of g (hence $1{\cdot}1$ and $1{\cdot}2$); position 2 designates the second argument of f (here z, a variable, hence with no further extension).



Definition (*Replacement at a position*). Let $t \in \mathcal{T}(\Sigma,\mathcal{X})$, $p \in \mathrm{Pos}(t)$, and $r \in \mathcal{T}(\Sigma,\mathcal{X})$. The term $t[r]_p$ (t with its subterm at position p replaced by r) is defined by induction on p:

$$t[r]_\varepsilon = r, \qquad\qquad f(t_1,\dots,t_n)[r]_{i\cdot p} = f(t_1,\dots,\ t_i[r]_p,\ \dots,t_n).$$

By construction, $(t[r]_p)|_p = r$: replacing at p and then reading back at p recovers exactly r.

Example. With $t = f(g(x,y),z)$ as before, and $r = h(0)$:

$$t[r]_1 = f(h(0),z), \qquad t[r]_2 = f(g(x,y),h(0)), \qquad t[r]_{1\cdot1} = f(g(h(0),y),z).$$

Nota.

- For $t \in \mathcal{T}(\Sigma,\mathcal{X})$ not a variable, $\mathrm{root}(t)$ denotes the outermost function symbol of t: if $t = f(t_1,\dots,t_n)$, then $\mathrm{root}(t) = f$.
- For p,q two positions, we write $p \le q$ if p is a *prefix* of q, i.e. there exists p' such that $q = p \cdot p'$; in this case $t|_q = (t|_p)|_{p'}$.
- We write $p < q$ if $p \le q$ and $p \ne q$ (*strict* prefix): the subterm at q occurs strictly inside the subterm at p.
- We write $p \parallel q$ if p and q are *disjoint* (or *incomparable*), i.e. neither $p \le q$ nor $q \le p$. In this case, for any term t with $p,q \in \mathrm{Pos}(t)$ and any term r, we have $(t[r]_p)|_q = t|_q$ (replacing at p does not affect a disjoint position q).
- For two positions $p,q \in (\mathbb{N}_{>0})^*$, exactly one of the following holds:

$$p = q, \qquad p < q, \qquad q < p, \qquad p \parallel q.$$

Note. Terms are naturally pictured as ordered trees: a leaf for each variable or constant, and a node labeled f with n ordered children for each subterm $f(t_1,\dots,t_n)$; the position $p \in \mathrm{Pos}(t)$ then designates a node of this tree, and $t|_p$ the subtree rooted there. This picture is purely informal and is never used directly in proofs, which always argue on positions and substitutions. In a similar spirit, we freely use *triangle* (or *diamond*) *diagrams* throughout this paper, e.g. a peak $t \;{}_R\!\leftarrow s \to_R u$ joined by a common reduct w, as an informal visual aid alongside the formal argument; these diagrams carry no formal content beyond what is proved in the surrounding text.

Definition (*strict subterm order*). We write $s \sqsubset t$ if there exists $p \in \mathrm{Pos}(t)$, $p \neq \varepsilon$, with $s = t|_p$; and $s \sqsubseteq t$ if $s \sqsubset t$ or $s = t$.

Lemma (*well-foundedness of the strict subterm order*). $\sqsubset$ is well-founded on $\mathcal{T}(\Sigma, \mathcal{X})$.

Proof. Let $|t|$ count the total number of symbols of t (defined by structural recursion, which terminates since every term is built in finitely many steps). If $s \sqsubset t$ via a position of length 1, an immediate computation gives $|s| < |t|$; by induction on the length of the position, $|s| < |t|$ for every $s \sqsubset t$. An infinite $\sqsubset$-decreasing chain would give an infinite strictly decreasing sequence of positive integers, impossible. $\square$

Definition (*Redex*). Let R be a TRS, $t \in \mathcal{T}(\Sigma, \mathcal{X})$, and $p \in \mathrm{Pos}(t)$. We say $t|_p$ is a *redex* (of R) if there exist a rule $l \to r \in R$ and a substitution σ such that $t|_p = l\sigma$.

Definition (*Rewrite step*). Let R be a TRS, $t \in \mathcal{T}(\Sigma, \mathcal{X})$, $p \in \mathrm{Pos}(t)$ such that $t|_p$ is a redex via rule $\ell \to r \in R$ and substitution σ. The *rewrite step at position p using $\ell \to r$* replaces $t|_p$ by $r\sigma$ inside t:

$$t \xrightarrow[p]{\ell \to r} t[r\sigma]_p.$$

We write $t \to_R t'$ if $t' = t[r\sigma]_p$ for some rule $\ell \to r \in R$, position $p \in \mathrm{Pos}(t)$, and substitution σ with $t|_p = \ell\sigma$.

Example. Let $\Sigma = \{0, s, +\}$ and $R = \{\, +(0, y) \to y, \quad +(s(x), y) \to s(+(x, y)) \,\}$. Consider $t = +(s(0), s(0))$.

- At $p = \varepsilon$: $t|_\varepsilon = t = +(s(0), s(0))$ is a redex via the second rule, with $\sigma(x) = 0$, $\sigma(y) = s(0)$. So $t \xrightarrow[\varepsilon]{+(s(x),y) \to s(+(x,y))} s(+(0, s(0)))$.
- In the resulting term $s(+(0, s(0)))$, the subterm at position 1 is $+(0, s(0))$, a redex via the first rule with $\sigma(y) = s(0)$: $s(+(0, s(0))) \xrightarrow[1]{+(0,y) \to y} s(s(0))$.

So $t \to_R^+ s(s(0))$ (in two steps), and $s(s(0))$ is a normal form: no subterm of $s(s(0))$ matches a left-hand side of R.

Definition (*Iterated rewriting*). Let R be a TRS and $a, b \in \mathcal{T}(\Sigma, \mathcal{X})$. For $n \in \mathbb{N}$, we define $a \to_R^n b$ ("a rewrites to b in exactly n steps") by induction on n:

$$a \to_R^0 b \iff a = b, \qquad a \to_R^{n+1} b \iff \exists\, c,\ a \to_R c \text{ and } c \to_R^n b.$$

We further write

$$a \to_R^* b \iff \exists\, n \in \mathbb{N},\ a \to_R^n b, \qquad a \to_R^+ b \iff \exists\, n \in \mathbb{N}^*,\ a \to_R^n b.$$

Convention. Throughout this paper, a reduction $a \to_R^* b$ may have length 0 (i.e. $a = b$). Several proofs below single out the "first step" of a reduction $a \to_R^* b$ by writing $a \to_R b_1 \to_R^* b$; this presupposes the reduction has length ≥ 1. The degenerate case $a = b$ is always handled trivially, by an immediate substitution (typically taking the sought witness to be b itself, or noting that b is joinable with anything a is joinable with), and we do not restate this at each occurrence below.

> **Definition** (*Conversion*). Let R be a TRS. We write $a \leftrightarrow_R b$ if $a \to_R b$ or $b \to_R a$. For $n \in \mathbb{N}$, $a \leftrightarrow_R^n b$ is defined by induction exactly as above, with $\leftrightarrow_R$ in place of $\to_R$. We write $a \leftrightarrow_R^* b$ if $\exists n,\ a \leftrightarrow_R^n b$: this is the smallest equivalence relation containing $\to_R$, called *conversion*.

Note. $\leftrightarrow_R^*$ is exactly the smallest congruence containing $\to_R$ and stable under substitution and context; it coincides with $=_E$ when R is obtained by orienting every equation of E (in either direction).

> **Definition** (*Joinable*). Two terms a and b are joinable if there exists a term c such that $a \to_R^* c$ and $b \to_R^* c$; we write it $a \downarrow_R b$.

> **Definition** (*Pre-diamond*). If a, b, c are three terms such that $b \leftarrow a \to c$, the triple (a, b, c) is called a *pre-diamond*, and a is its *root*. A pre-diamond is *proper* if a, b, c are pairwise distinct. A pre-diamond (a, b, c) is *joinable* if the pair (b, c) is joinable.

Note. A non-proper pre-diamond is always joinable.

> **Definition** (*normal form*). Let R be a term rewriting system over $\mathcal{T}(\Sigma, \mathcal{X})$. A term t is said to be a *normal form* (or *irreducible*) for R if there is no term t' such that $t \to_R t'$.

> **Definition** (*Term with a normal form*). Let R be a TRS and $u \in \mathcal{T}(\Sigma, \mathcal{X})$. We say that *$u$ has a normal form* if there exists a normal form $t \in \mathcal{T}(\Sigma, \mathcal{X})$ for R such that $u \leftrightarrow_R^* t$.

> **Definition** (*Weak normalization*). A TRS R is *weakly normalizing* (WN) if every term $t \in \mathcal{T}(\Sigma, \mathcal{X})$ has a normal form.

3 Core Properties

3.1 Confluence and Termination

> **Definition** (*Confluence*). A TRS R is *confluent* if, for all terms $s, t, u \in \mathcal{T}(\Sigma, \mathcal{X})$ with $s \to_R^* t$ and $s \to_R^* u$, there exists v such that $t \to_R^* v$ and $u \to_R^* v$.

$$
\begin{tikzcd}
 & t \arrow[dr, dashed, "*"] & \\
s \arrow[ur, "*"] \arrow[dr, "*"'] & & v \\
 & u \arrow[ur, dashed, "*"'] &
\end{tikzcd}
$$

Remark. This definition can be read through the lens of equational theories: if R orients every equation of an equational theory E, confluence of R is precisely what guarantees that joinability by $\to_R^*$ coincides with (and thus effectively computes) equality in $=_E$; see the discussion of the word problem below.

Definition (*Local confluence*). A TRS R is *locally confluent* if for all $t, u, v \in \mathcal{T}(\Sigma, \mathcal{X})$ with $t \to_R u$ and $t \to_R v$, there exists h such that $u \to_R^* h$ and $v \to_R^* h$.

$$
\begin{tikzcd}
 & u \arrow[dr, dashed, "*"] & \\
t \arrow[ur] \arrow[dr] & & h \\
 & v \arrow[ur, dashed, "*"'] &
\end{tikzcd}
$$

Equivalently, a TRS R is locally confluent iff every pre-diamond is joinable: the only difference with confluence above is that the two initial arrows are single rewrite steps rather than arbitrary reductions.

Definition (*Church–Rosser property*). A TRS R is called *Church–Rosser* if for all $a, b \in \mathcal{T}(\Sigma, \mathcal{X})$ such that $a \leftrightarrow_R^* b$, there exists v such that $a \to_R^* v$ and $b \to_R^* v$.

Definition (*Termination*). $\to_R$ is *terminating* if every rewrite sequence $t_0 \to_R t_1 \to_R t_2 \to_R \cdots$ is finite.

Definition (*Noetherian order*). A strict ordering is *Noetherian* (or *well-founded*) if every non-empty set has a minimal element.

Proposition (*equivalence of well-foundedness formulations*). A strict order $\prec$ on a set A is Noetherian if and only if there is no infinite $\prec$-decreasing sequence $a_0 \succ a_1 \succ a_2 \succ \cdots$ in A.

Proof. ($\Rightarrow$) Suppose such an infinite sequence exists. The set $\{a_k : k \in \mathbb{N}\}$ is non-empty, hence has a $\prec$-minimal element a_{k_0}; but $a_{k_0+1} \prec a_{k_0}$ with a_{k_0+1} in the same set, contradicting minimality of a_{k_0}.

($\Leftarrow$) Suppose some non-empty $B \subseteq A$ has no $\prec$-minimal element. Fix $b_0 \in B$. Since b_0 is not minimal in B, there exists $b_1 \in B$ with $b_1 \prec b_0$. By induction, having built $b_0, \ldots, b_n \in B$ with $b_n \prec \cdots \prec b_0$, since b_n is not minimal in B there exists $b_{n+1} \in B$ with $b_{n+1} \prec b_n$ (selecting one such witness at each of the countably many steps of this construction invokes the axiom of dependent choice). This produces an infinite $\prec$-decreasing sequence in $B \subseteq A$, contradicting the hypothesis. $\square$

Nota. This equivalence is used freely throughout the paper: some proofs establish well-foundedness via the absence of an infinite decreasing sequence (e.g. the lemma of well-foundedness of the strict subterm order, Section 2), while others extract a minimal element directly to argue by contradiction (e.g. the proof of properties of $\prec_R$ below, or the local-confluence lemma underlying the Critical Pair Theorem). The proposition above licenses moving freely between the two.

Nota (on the axiom of choice). The ($\Leftarrow$) direction of the proposition above is not a theorem of set theory without some choice principle: "no infinite decreasing sequence" is, in general, strictly weaker than "every non-empty subset has a minimal element" unless at least the axiom of *dependent choice* (DC) is assumed: a standard, mild principle (strictly weaker than full AC, and free of its more counter-intuitive consequences) that we take for granted here, as is universal practice in this field. That said, in the two concrete instances used in this paper, the dependency on choice can in fact be avoided by a more direct argument: for the strict subterm order $\sqsubset$, the term-size measure $|t| \in \mathbb{N}$ gives, for any non-empty A, a minimal element directly by well-ordering of $\mathbb{N}$ (itself choice-free): take any $a_0 \in A$ achieving the least value of $|\cdot|$ on A; no $b \in A$ can satisfy $b \sqsubset a_0$, since that would force $|b| < |a_0|$, contradicting minimality of $|a_0|$ among $\{|a| : a \in A\}$. This is a single existential choice, not an infinite sequence of dependent ones. The order $\prec_R$ below, by contrast, is defined purely in terms of $\to_R^+$ with no such built-in numerical measure supplied by termination alone, and its uses of "minimal element" genuinely rest on the proposition above (hence on DC) in general.

Definition (*Ordering relation associated with a TRS*). Given a terminating TRS R on $\mathcal{T}(\Sigma, \mathcal{X})$, we denote $\prec_R$ the relation:

$$\forall a, b \in \mathcal{T}(\Sigma, \mathcal{X}), \quad b \prec_R a \iff a \to_R^+ b.$$

Proposition (*Properties of* $\prec_R$).

1. Since $\to_R^+$ is the transitive closure of $\to_R$, it is trivial that $\prec_R$ is a strict pre-order.
2. Since R terminates, $\prec_R$ is furthermore antisymmetric, and thus a strict ordering relation.
3. Again since R terminates, $\prec_R$ is a Noetherian ordering.

Proof. 1. $\prec_R$ is transitive. Let $a, b, c \in \mathcal{T}(\Sigma, \mathcal{X})$ with $c \prec_R b$ and $b \prec_R a$, i.e. $b \to_R^+ c$ and $a \to_R^+ b$. Since $\to_R^+$ is the transitive closure of $\to_R$, $a \to_R^+ b$ and $b \to_R^+ c$ give $a \to_R^+ c$, i.e. $c \prec_R a$. Hence $\prec_R$ is transitive; it is a strict pre-order (irreflexivity being immediate, since R terminates: $a \to_R^+ a$ would mean some finite non-empty sequence of steps returns from a to a; concatenating this cycle with itself indefinitely, $a \to_R \cdots \to_R a \to_R \cdots \to_R a \to_R \cdots$, yields an infinite reduction, contradicting termination).

Remark. Since $\prec_R$ is irreflexive and transitive (point 1), antisymmetry (point 2 below) already follows for free, without any further appeal to termination: if $a \prec_R b$ and $b \prec_R a$, transitivity gives $a \prec_R a$, contradicting irreflexivity. We nonetheless give a direct, self-contained argument below, phrased in terms of reductions rather than of the abstract order, as it makes the underlying mechanism (concatenating two witnessing cycles into one infinite reduction) explicit.

2. $\prec_R$ is antisymmetric. Suppose, for contradiction, that $a \prec_R b$ and $b \prec_R a$ for some $a \neq b$. This means $b \to_R^+ a$ and $a \to_R^+ b$. Concatenating a witnessing sequence of steps for each, we obtain an infinite reduction $a \to_R \cdots \to_R b \to_R \cdots \to_R a \to_R \cdots$, contradicting termination of R. Hence $\prec_R$ is antisymmetric, and therefore a strict ordering relation.

3. $\prec_R$ is Noetherian. By the proposition on equivalence of well-foundedness formulations, it suffices to show there is no infinite $\prec_R$-decreasing sequence. Suppose, for contradiction, that $t_0 \succ_R t_1 \succ_R t_2 \succ_R \cdots$ is such a sequence. By definition of $\prec_R$, each $t_k \succ_R t_{k+1}$ means $t_k \to_R^+ t_{k+1}$, i.e. there exists a finite non-empty sequence of rewrite steps from t_k to t_{k+1}. Concatenating these sequences for every k yields an infinite reduction sequence issued from t_0, contradicting termination of R. Hence no infinite $\prec_R$-decreasing sequence exists, and by the proposition on equivalence of well-foundedness formulations, $\prec_R$ is Noetherian. $\square$

Definition (*Complete*). R is *complete*, or *convergent*, if R is terminating and confluent.

Remark. If R is convergent and orients an equational theory E (i.e. $\leftrightarrow_R^* = =_E$), then for all $t, t' \in \mathcal{T}(\Sigma, \mathcal{X})$:

$$t =_E t' \iff \exists\, u,\ t \to_R^* u \text{ and } t' \to_R^* u.$$

Indeed, $t =_E t'$ means $t \leftrightarrow_R^* t'$; since R is confluent, this conversion yields a common reduct u by joinability. This is precisely what makes the word problem for E decidable once R is convergent: compute the (unique) normal forms of t and t' and compare them syntactically.

Proposition (*Canonical representatives via convergence*). Let E be an equational theory and R a convergent TRS orienting E (i.e. $\leftrightarrow_R^* = =_E$). Then the map sending each equivalence class $[t]_{=_E} \in \mathcal{T}(\Sigma, \mathcal{X})/{=_E}$ to the (unique) normal form of any of its representatives is well defined, and gives a bijection

$$\mathcal{T}(\Sigma, \mathcal{X})/{=_E} \xrightarrow{\sim} \{\text{normal forms of } R\}.$$

Proof. Well defined. Let $t, t' \in [u]_{=_E}$, i.e. $t \leftrightarrow_R^* t'$. Since R is confluent, t and t' have a common reduct w; since R terminates, t and t' each reduce to some normal form, and by confluence these normal forms coincide (a normal form cannot reduce further, so w itself must reduce to, and hence equal, the normal form of both t and t'). Hence the normal form does not depend on the choice of representative within the class.

Injective. If two classes have the same normal form $\bar{u}$, then $t \to_R^* \bar{u} \;{}^*\!\!\leftarrow_R t'$, so $t \leftrightarrow_R^* t'$: the two classes coincide.

Surjective. Every normal form $\bar{u}$ is trivially the normal form of $[\bar{u}]_{=_E}$. $\square$

3.2 Newman's Lemma

Newman's Lemma is the central bridge between the *local* property of confluence (checkable, in principle, redex by redex) and the *global* property of confluence (joinability of arbitrarily long derivations). We give two independent proofs below: the first proceeds by structural induction along the well-founded order $\prec_R$, directly establishing the desired joinability property; the second argues by contradiction on a minimal counterexample. Both are standard in the literature, and we present them side by side because they illustrate two complementary proof techniques (direct well-founded induction versus minimal-counterexample) that recur throughout this paper; in particular, the second technique is exactly the one used later for the Critical Pair Theorem and for the modularity of confluence.

> **Theorem** (*Newman's Lemma*). A *locally confluent* and *terminating* term rewriting system is confluent.

$$\begin{tikzcd}
a \arrow[r] \arrow[d] & b_1 \arrow[r, "*"] \arrow[d, "*"] & b \arrow[d, "*"] \\
c_1 \arrow[r, "*"] \arrow[d, "*"] & u \arrow[r, "*"] \arrow[d, "*"] & v \arrow[d, "*"] \\
c \arrow[r, "*"] & e \arrow[r, "*"] & s
\end{tikzcd}$$

Proof 1 (well-founded induction). Since R is terminating, the relation $\to_R$ is well-founded, and thus induces the order relation $\prec_R$ of the previous section. The diagram above summarises the construction carried out below: starting from the peak a, local confluence at the first step produces u, and three successive applications of the induction hypothesis P (at c_1, at b_1, and at u) chase the diagram down to a common point s.

The fact that R terminates allows us to carry out a proof by induction to show the property P defined by

$$\forall a \in \mathcal{T}(\Sigma, \mathcal{X}), \quad P(a): \ \forall b, c \text{ with } a \to_R^* b \text{ and } a \to_R^* c,\ \exists\, d,\ b \to_R^* d \text{ and } c \to_R^* d.$$

Base case. The base case concerns terms a that do not rewrite to any other term. Let such an a be given. For b, c such that $a \to_R^* b$ and $a \to_R^* c$, we can only have $c = b = a$. So $d = a$ gives us $b \to_R^* d$ and $c \to_R^* d$. Hence $P(a)$.

Inductive case. Let a; assume that for all $a' \prec_R a$, $P(a')$ holds. Let b, c be such that $a \to_R^* c$ and $a \to_R^* b$. If $b = a$ or $c = a$, $P(a)$ holds trivially (by the Convention above: e.g. if $b = a$, take $d := c$, since $b = a \to_R^* c = d$ and $c \to_R^* d$ trivially). Assume henceforth $b \neq a$, $c \neq a$: both reductions have length ≥ 1, so we may write $a \to_R c_1 \to_R^* c$ and $a \to_R b_1 \to_R^* b$.

By local confluence, there exists u such that $c_1 \to_R^* u$ and $b_1 \to_R^* u$.

Since $c_1 \to_R^* u$ and $c_1 \to_R^* c$, and $c_1 \prec_R a$ (as $a \to_R c_1$ is a single step, hence $a \to_R^+ c_1$), $P(c_1)$ holds: there exists e such that $u \to_R^* e$ and $c \to_R^* e$.

Since $b_1 \to_R^* b$ and $b_1 \to_R^* u$, and $b_1 \prec_R a$ (likewise, from $a \to_R b_1$), $P(b_1)$ holds: there exists v such that $b \to_R^* v$ and $u \to_R^* v$.

We now have $u \to_R^* e$ and $u \to_R^* v$, and $u \prec_R a$ (since $a \to_R c_1 \to_R^* u$ gives $a \to_R^+ u$), so $P(u)$ holds: there exists s such that $e \to_R^* s$ and $v \to_R^* s$. Then $b \to_R^* v \to_R^* s$ and $c \to_R^* e \to_R^* s$: the peak b, c is joinable, which concludes the proof. $\square$

Proof 2 (minimal counterexample). Let R be a terminating and locally confluent TRS. Let

$$A = \{a \in \mathcal{T}(\Sigma, \mathcal{X}) \mid \exists\, c, b : c \leftarrow_R^+ a \to_R^+ b,\ \nexists\, d,\ b \to_R^* d \leftarrow_R^* c\}.$$

(We use $\to_R^+$, rather than $\to_R^*$ as in the definition of confluence, because a pair involving a length-0 reduction is always trivially joinable by the Convention above: such a pair can never witness a failure of confluence, so it is harmless to exclude it from A.)

We prove that A is empty by contradiction. Assume $A \neq \varnothing$. Let a be a minimal element of A for $\prec_R$; such an element exists since $\prec_R$ is Noetherian.

By definition of A, fix witnessing terms b, c with $a \to_R^+ b$, $a \to_R^+ c$, not joinable.

If both witnessing reductions consist of a single rewriting step, then, by local confluence, there exists d such that $b \to_R^* d$ and $c \to_R^* d$: b, c are joinable, contradicting $a \in A$. (This case is in fact a special instance of the general argument below, with $b_1 = b$, $c_1 = c$; we single it out only for clarity.)

In general, since both witnessing reductions have length ≥ 1, there exist b_1, c_1 such that $b \leftarrow_R^* b_1 \leftarrow_R a \to_R c_1 \to_R^* c$.

Since $a \to_R b_1$ and $a \to_R c_1$ are single steps, local confluence yields e with $b_1 \to_R^* e$ and $c_1 \to_R^* e$.

Since $a \to_R c_1$ is a single step, $c_1 \prec_R a$; minimality of a in A gives $c_1 \notin A$, so the pair (c, e) (both reachable from c_1) cannot witness a failure of confluence at c_1: c and e are joinable in some u (if either $c_1 \to_R^* c$ or $c_1 \to_R^* e$ happens to have length 0, this is immediate directly from $c_1 \to_R^* e$ or $c_1 \to_R^* c$; otherwise it follows from $c_1 \notin A$, by the argument just given).

Similarly, $a \to_R b_1$ gives $b_1 \prec_R a$, hence $b_1 \notin A$: b and e are joinable in some f, by the same reasoning.

Finally, since $a \to_R c_1 \to_R^* e$, we have $a \to_R^+ e$, i.e. $e \prec_R a$, hence $e \notin A$: u and f are joinable in some g. It follows that b and c are joinable, contradicting $a \in A$.

Therefore A is empty: every terminating and locally confluent rewrite system is confluent. □

> **Theorem** (*Confluence and termination*). A terminating TRS is confluent if and only if it is locally confluent.

Proof. Newman's Lemma already establishes that a terminating and locally confluent TRS is confluent. The converse (a terminating confluent TRS is locally confluent) is immediate, since local confluence is a particular case of confluence. □

> **Lemma** (*confluence implies joinability of conversions*). If R is confluent, then for all terms a, b such that $a \leftrightarrow_R^* b$, there exists a term c such that $a \to_R^* c \leftarrow_R^* b$.

$$\begin{tikzcd}
a_0 \arrow[r, leftrightarrow, dashed, "\text{conversion, length } n+1"] \arrow[d, "*"'] & a_{n+1} \arrow[d, "*"] \\
e \arrow[r, "*"'] & u
\end{tikzcd}$$

Intuitively, the induction shows that a zigzag conversion of any length can always be flattened into a single valley: the picture above illustrates the inductive step, replacing a conversion of length $n + 1$ by a common reduct u built from the shorter conversion's own common reduct e.

Proof. Let $P(n)$ be: "for every conversion $a \leftrightarrow_R^n b$ of exactly n links, there exists c with $a \to_R^* c \leftarrow_R^* b$" (using the iterated conversion relation $\leftrightarrow_R^n$ of the definition of conversion). Since $a \leftrightarrow_R^* b$ means $a \leftrightarrow_R^n b$ for some $n \in \mathbb{N}$, establishing $P(n)$ for every n proves the lemma.

Base case $(n = 0)$**.** If $a \leftrightarrow_R^0 b$, then $a = b$; take $c = a$.

Induction step. Assume $P(n)$. Consider a conversion of length $n+1$, written with $n+2$ terms $a_0 \leftrightarrow_R a_1 \leftrightarrow_R \cdots \leftrightarrow_R a_{n+1}$ (so that the prefix $a_0 \leftrightarrow_R \cdots \leftrightarrow_R a_n$ has exactly n links). By the induction hypothesis applied to this prefix, there exists e with $a_0 \to_R^* e$ and $a_n \to_R^* e$. If $a_n \to_R a_{n+1}$, then, since $a_n \to_R^* e$ and $a_n \to_R a_{n+1}$ (hence $a_n \to_R^* a_{n+1}$), confluence gives u with $e \to_R^* u$ and $a_{n+1} \to_R^* u$; combined with $a_0 \to_R^* e$, this gives $a_0 \to_R^* u \leftarrow_R^* a_{n+1}$. If $a_{n+1} \to_R a_n$, then $a_{n+1} \to_R a_n \to_R^* e$, so $a_{n+1} \to_R^* e$, together with $a_0 \to_R^* e$, gives $a_0 \to_R^* e \leftarrow_R^* a_{n+1}$. In both cases $P(n+1)$ holds, taking $c := u$ or $c := e$ respectively. ☐

Theorem (*Termination and Church–Rosser*). A terminating TRS R is Church–Rosser if and only if it is locally confluent (equivalently, confluent).

Proof. The previous lemma establishes Confluence ⇒ Church–Rosser. We now prove the converse. Assume R satisfies Church–Rosser. Let a, b, c be such that $a \to_R^* b$ and $a \to_R^* c$. Then $c \leftarrow_R^* a \to_R^* b$, hence $c \leftrightarrow_R^* a \leftrightarrow_R^* b$ (using $\to_R^* \subseteq \leftrightarrow_R^*$, immediate from $\to_R \subseteq \leftrightarrow_R$ by monotonicity of reflexive-transitive closure), so $c \leftrightarrow_R^* b$. By Church–Rosser, there exists d with $c \to_R^* d$ and $b \to_R^* d$. Hence every pair of reductions from the same term is joinable, so R is confluent. ☐

Remark. Termination is in fact not used anywhere in the proof above: the equivalence Church–Rosser ⇔ confluent holds for an arbitrary TRS, with no termination assumption whatsoever. Termination is only needed for the parenthetical "(equivalently, confluent)", which further identifies confluence with local confluence via the theorem on confluence and termination proved just before.

4 Unification

The Critical Pair Theorem, our next goal, will rely crucially on the existence of a *most general unifier* between two patterns: this is precisely how one detects, and resolves, the ambiguities that threaten local confluence. We therefore develop this notion now: we present the classical rule-based unification algorithm, prove its termination, and establish that it computes a most general unifier whenever one exists. This ensures that Section 5 rests on solid ground, rather than assuming unifiability as a black box.

Nota (composition of substitutions). For substitutions σ, θ, we write $\sigma \circ \theta$ for their composition, defined so as to match the postfix notation $x\sigma$ used throughout: $x(\sigma \circ \theta) := (x\sigma)\theta$ for every variable x (apply σ first, then θ). This extends to all terms by the usual homomorphic extension of substitutions.

Definition (*Unifier*). A substitution σ is a *unifier* of two terms s, t if $s\sigma = t\sigma$. A *most general unifier* (MGU) of s, t is a unifier σ such that for every unifier τ of s, t there exists a substitution θ with $\tau = \sigma \circ \theta$.

Definition (*Unification problem*). A *unification problem* is a finite set of equations $P = \{s_1 \doteq t_1, \ldots, s_k \doteq t_k\}$ with $s_i, t_i \in \mathcal{T}(\Sigma, \mathcal{X})$. A substitution σ *solves* P if $s_i\sigma = t_i\sigma$ for every i. We write $\mathrm{Sol}(P)$ for the set of unifiers of P.

Definition (*Unification transition relation*). We write $P \vdash P'$ (read: "P transitions to P'"), for two unification problems P, P' (or $P' = \textsc{Fail}$, a distinguished failure symbol), to mean that P' is obtained from P by one application of one of the six inference rules of Definition [Unification algorithm] below. We write $\vdash^*$ for the reflexive-transitive closure of $\vdash$. This symbol is overloaded in the literature: in proof theory it usually denotes syntactic derivability. Here, and in the analogous relation used later for Knuth–Bendix completion, it denotes exclusively a single step of a transition system on syntactic configurations, unrelated to logical derivability.

Definition (*Unification algorithm*). The unification algorithm is a transition system on unification problems (together with an accumulated substitution, initially the identity), governed by the following rules, applied to an arbitrary equation $s \doteq t \in P$ chosen non-deterministically:

Delete. $\{s \doteq s\} \cup P \ \vdash \ P$.

Decompose. $\{f(s_1, \dots, s_n) \doteq f(t_1, \dots, t_n)\} \cup P \ \vdash \ \{s_1 \doteq t_1, \dots, s_n \doteq t_n\} \cup P$.

Clash. $\{f(s_1, \dots, s_n) \doteq g(t_1, \dots, t_m)\} \cup P \ \vdash \ \textsc{Fail}$, if $f \neq g$.

Orient. $\{t \doteq x\} \cup P \ \vdash \ \{x \doteq t\} \cup P$, if $t \notin \mathcal{X}$ and $x \in \mathcal{X}$.

Eliminate. $\{x \doteq t\} \cup P \ \vdash \ \{x \doteq t\} \cup P[x := t]$, if $x \in \mathcal{X}$, $x \notin \mathrm{Var}(t)$, and x occurs elsewhere in P (here $P[x := t]$ denotes P with every other occurrence of x replaced by t).

Occurs-check. $\{x \doteq t\} \cup P \ \vdash \ \textsc{Fail}$, if $x \in \mathcal{X}$, $x \neq t$, and $x \in \mathrm{Var}(t)$.

A problem P is in *solved form* if every equation is of the shape $x \doteq t$ with $x \in \mathcal{X}$ occurring exactly once in the whole of P (i.e. nowhere else, neither on a left- nor a right-hand side of another equation), in which case it directly defines a substitution $\sigma_P(x) := t$ (and $\sigma_P(y) := y$ for y not occurring on the left of an equation of P).

Lemma (*termination of the unification algorithm*). No infinite chain of transitions $P_0 \vdash P_1 \vdash P_2 \vdash \cdots$ (other than immediately reaching $\textsc{Fail}$ or a solved form) exists.

Proof. Define a measure $\mu(P) = (n_1(P),\ n_2(P),\ n_0(P)) \in \mathbb{N} \times \mathbb{N} \times \mathbb{N}$, ordered *lexicographically in this order* (so n_1 is checked first, then n_2, and n_0 only as a tie-breaker), where:

- $n_1(P)$ is the number of distinct variables occurring in P that are *not* the left-hand side of a solved equation;
- $n_2(P)$ is the total number of symbols occurring in P (summing the sizes of all s_i, t_i);
- $n_0(P)$ is the number of equations of P of the form $t \doteq x$ with $t \notin \mathcal{X}$, $x \in \mathcal{X}$ (i.e. not yet oriented).

Delete does not increase n_1 (no variable is introduced or eliminated) and strictly decreases n_2 (one equation, hence at least one symbol, is removed). Since n_1 does not increase, this alone secures a strict lexicographic decrease of μ, regardless of how n_0 behaves.

Decompose replaces one equation between two terms of size ≥ 2 by $n \geq 1$ equations between their immediate arguments: n_1 is unchanged (no variable occurrence is created or destroyed, only redistributed among the new equations), while n_2 strictly decreases (the n occurrences of the shared root symbol f are removed). As with Delete, this secures a strict decrease of μ at the second component, regardless of n_0. *We stress that n_0 may in fact strictly* increase *under Decompose*: e.g. decomposing $f(g(y), a) \doteq f(z, a)$ produces $g(y) \doteq z$, a new equation of the "unoriented" shape $t \doteq x$ that did not exist before. This is harmless precisely because

n_0 is the *least* significant component of μ: the strict decrease already secured at n_2 (a more significant position) is unaffected by any change to n_0.

Orient leaves n_1, n_2 exactly unchanged (it only swaps the two sides of a single equation), so it is exactly the case where the tie-breaker n_0 must do the work: Orient strictly decreases n_0, since it converts one equation of the form $t \doteq x$ (counted in n_0) into $x \doteq t$ (with $x \in \mathcal{X}$ now on the left, so it no longer matches this shape, and cannot be re-oriented, hence permanently excluded from future counts of n_0), without affecting the orientation status of any other equation of P. Hence μ strictly decreases lexicographically (first two components tied, third strictly decreases).

Eliminate strictly decreases n_1: the variable x no longer occurs in P except as the (now unique) left-hand side of $x \doteq t$, since every other occurrence has been substituted away; no new variable is introduced (t's variables were already present, as $P[x := t]$ only replaces occurrences of x, which does not introduce fresh variables). Since n_1 is the most significant component, this alone secures a strict decrease of μ, regardless of how n_2 or n_0 are affected by the substitution $P[x := t]$ (which may in fact increase both, e.g. if x occurred several times and t is large).

Clash and **Occurs-check** terminate the process immediately (Fail).

Since μ takes values in $\mathbb{N}^3$ with the lexicographic order (a well-founded order, being the lexicographic product of well-founded orders) and strictly decreases at every non-terminal transition, no infinite chain of transitions exists. □

Theorem (*Correctness of unification*). Let P_0 be a unification problem. Every maximal run of the unification algorithm from P_0 terminates either in Fail, or in a solved form P defining a substitution σ_P. In the first case, $\mathrm{Sol}(P_0) = \emptyset$. In the second, σ_P is a most general unifier of P_0.

Proof. Termination of every run is Lemma [termination of the unification algorithm]. We show that every transition rule preserves the set of solutions, i.e. if $P \vdash P'$ (with $P' \neq \text{Fail}$) then $\mathrm{Sol}(P) = \mathrm{Sol}(P')$, and that Clash/Occurs-check fire exactly when $\mathrm{Sol}(P) = \emptyset$.

Delete. σ solves $\{s \doteq s\} \cup P$ iff $s\sigma = s\sigma$ (always true) and σ solves P: same solution set.

Decompose. σ solves $f(s_1, \ldots, s_n) \doteq f(t_1, \ldots, t_n)$ iff $f(s_1, \ldots, s_n)\sigma = f(t_1, \ldots, t_n)\sigma$ iff $s_i\sigma = t_i\sigma$ for every i (by injectivity of term formation: two terms with the same root and the same arguments, position by position, are equal, and conversely): same solution set as the decomposed system.

Clash. If $f \neq g$, no substitution can make $f(\ldots)\sigma = g(\ldots)\sigma$, since substitution never changes the root symbol of a non-variable term: $f(\ldots)\sigma$ is rooted by f, $g(\ldots)\sigma$ by g. Hence $\mathrm{Sol}(P) = \emptyset$.

Orient. σ solves $t \doteq x$ iff $t\sigma = x\sigma$ iff $x\sigma = t\sigma$: same solution set as $x \doteq t$.

Eliminate. Suppose σ solves $\{x \doteq t\} \cup P$: then $x\sigma = t\sigma$, and for any equation $u \doteq v \in P$, $(u[x := t])\sigma = u\sigma[x := t\sigma\text{-substituted for } x\sigma] = u\sigma$ (since substituting x by t inside u and then applying σ is the same as applying σ directly, using $x\sigma = t\sigma$), so σ also solves $P[x := t]$, hence $\{x \doteq t\} \cup P[x := t]$. Conversely if σ solves $\{x \doteq t\} \cup P[x := t]$, then since $x\sigma = t\sigma$, for $u \doteq v \in P$, $u\sigma = (u[x := t])\sigma = v[x := t]\sigma$... (symmetric argument) $= v\sigma$: σ solves $u \doteq v$. Hence σ solves P, so $\{x \doteq t\} \cup P$. Same solution set.

Occurs-check. If $x \in \mathrm{Var}(t)$ and $x \neq t$, then t is (or contains as a proper subterm) a term strictly containing x; any solution would require $x\sigma = t\sigma$, but $t\sigma$ has strictly more symbols than $x\sigma$ whenever x occurs properly inside t (an elementary induction on the structure of

t, using that substitution only grows or preserves term size symbol by symbol), which is impossible for a fixed-point equation $x\sigma = t\sigma$ unless $x = t$. Hence $\mathrm{Sol}(\{x \doteq t\} \cup P) = \emptyset$.

Since every transition preserves Sol (or correctly detects $\mathrm{Sol} = \emptyset$), and $\mathrm{Sol}(P_0) = \mathrm{Sol}(P_{\text{final}})$ along any run, we conclude: if the run ends in FAIL, $\mathrm{Sol}(P_0) = \emptyset$. If it ends in a solved form P, then $\mathrm{Sol}(P_0) = \mathrm{Sol}(P)$, and σ_P (reading off $\sigma_P(x) = t$ for each solved equation $x \doteq t$) is well defined as a substitution, since each variable occurs at most once as a left-hand side of P. To see that σ_P solves P: for each solved equation $x \doteq t$, we have $x\sigma_P = t$ by definition; and $t\sigma_P = t$, because the solved-form condition (definition of unification algorithm: each left-hand side variable occurs *nowhere else* in P, neither as a left- nor as a right-hand side of another equation) forces every variable occurring in t to be one on which σ_P acts as the identity. Hence $x\sigma_P = t = t\sigma_P$: σ_P solves $x \doteq t$, for every solved equation of P, so $\sigma_P \in \mathrm{Sol}(P) = \mathrm{Sol}(P_0)$.

It remains to show σ_P is *most general*: let $\tau \in \mathrm{Sol}(P_0) = \mathrm{Sol}(P)$. We exhibit θ with $\tau = \sigma_P \circ \theta$ (in the sense of the Nota on composition of substitutions above) by simply taking $\theta := \tau$, and verifying $x(\sigma_P \circ \theta) = x\tau$ for every variable x: if x is a left-hand side of a solved equation $x \doteq t$ of P, then $x(\sigma_P \circ \theta) = (x\sigma_P)\theta = t\theta = t\tau$, and since τ solves $x \doteq t$, $t\tau = x\tau$; if x is not a left-hand side of P, $x\sigma_P = x$, so $x(\sigma_P \circ \theta) = (x\sigma_P)\theta = x\theta = x\tau$. In both cases $x(\sigma_P \circ \theta) = x\tau$, so $\sigma_P \circ \theta$ and τ agree on every variable, hence $\tau = \sigma_P \circ \theta$ (with $\theta = \tau$), witnessing that σ_P is a most general unifier. $\square$

Example. Unify $s = f(x, g(y))$ and $t = f(g(a), x)$ (with a a constant).

$$\begin{aligned}
\{f(x, g(y)) \doteq f(g(a), x)\} &\vdash_{\text{Decompose}} \{x \doteq g(a),\ g(y) \doteq x\} \\
&\vdash_{\text{Orient}} \{x \doteq g(a),\ x \doteq g(y)\} \\
&\vdash_{\text{Eliminate}} \{x \doteq g(a),\ g(a) \doteq g(y)\} \\
&\vdash_{\text{Decompose}} \{x \doteq g(a),\ a \doteq y\} \\
&\vdash_{\text{Orient}} \{x \doteq g(a),\ y \doteq a\}
\end{aligned}$$

This is a solved form, giving the most general unifier $\sigma(x) = g(a)$, $\sigma(y) = a$. One checks $s\sigma = f(g(a), g(a)) = t\sigma$.

5 Critical pairs

The property of confluence is very important for a TRS, especially for equational reasoning. It allows us to rewrite an expression in different ways and still obtain the same result. Generally, it is very difficult to obtain a confluent TRS directly. But in the special case of terminating TRSs, Newman's Lemma reduces the problem to ensuring local confluence. In this context, the objective becomes: how can we make a TRS locally confluent, in a way that is mechanically checkable?

Definition (*Overlap*). Let R be a term rewriting system. An *overlap* occurs when there exist rules $l_1 \to r_1$ and $l_2 \to r_2$ in R (sharing no variables, renamed apart if necessary), a non-variable position $p \in \mathrm{Pos}(l_1)$, and a substitution σ such that $l_1|_p\sigma = l_2\sigma$.

Definition (*Critical pair*). Let $l_1 \to_R r_1$ and $l_2 \to_R r_2$ be two rewrite rules sharing no variables (renamed apart if necessary). Suppose there exists a position $p \in \mathcal{P}os_{\mathcal{F}}(l_1)$ (i.e. p is not a variable position of l_1) and a most general unifier σ such that $l_1|_p\sigma = l_2\sigma$. This gives rise to two possible reductions of $l_1\sigma$: $l_1\sigma \to r_1\sigma$, and $l_1\sigma \to l_1\sigma[r_2\sigma]_p$. We call *critical pair* the pair $(r_1\sigma,\ l_1\sigma[r_2\sigma]_p)$. By convention, the trivial overlap of a rule with itself at the root position is excluded, as it always yields the trivial pair (r_1, r_1).

Algorithm 1 ComputeCriticalPairs

Require: A term rewriting system $R = \{l_1 \to_R r_1, \ldots, l_n \to_R r_n\}$
Ensure: The set $\mathrm{CP}(R)$ of all (non-trivial) critical pairs of R

function ComputeCriticalPairs(R)
    $CP \leftarrow \emptyset$
    for all rule $(l_1 \to_R r_1) \in R$ **do**
        for all rule $(l_2 \to_R r_2) \in R$ **do**
            $(l_2', r_2') \leftarrow \textsc{Rename}(l_2 \to_R r_2)$
            for all position $p \in \mathcal{P}os_{\mathcal{F}}(l_1)$ **do**
                if $(l_1 \to_R r_1) = (l_2 \to_R r_2)$ **and** $p = \varepsilon$ **then**
                    continue
                end if
                $\sigma \leftarrow \textsc{Unify}(l_1|_p,\ l_2')$
                if $\sigma \neq \textsc{Fail}$ **then**
                    $CP \leftarrow CP \cup \{(\sigma(r_1),\ \sigma(l_1[r_2']_p))\}$
                end if
            end for
        end for
    end for
    return CP
end function

Note. Let (s,t) be a critical pair of R. It is *trivial* if $s \downarrow_R t$. In particular every critical pair with $s = t$ is trivial.

Definition (*Parallel rewriting*). Let R be a TRS. For a term t, a *parallel rewrite step* is a relation $t \Rightarrow_R t'$ defined as follows: there exists a finite non-overlapping set of positions $P = \{p_1, \ldots, p_n\} \subseteq \mathrm{Pos}(t)$, and for each i a rule $\ell_i \to r_i \in R$ and substitution σ_i with $t|_{p_i} = \ell_i\sigma_i$, such that

$$t' = t[r_1\sigma_1]_{p_1} \cdots [r_n\sigma_n]_{p_n}.$$

Note. $\to_R \subseteq \Rightarrow_R \subseteq \to_R^*$.

Definition (*context*). Let $\{\Box_1, \ldots, \Box_n\}$ be fresh symbols, disjoint from $\Sigma \cup \mathcal{X}$, each treated as a constant. A *context* with holes $\Box_1, \ldots, \Box_n$ is a term $C \in \mathcal{T}(\Sigma,\ \mathcal{X} \cup \{\Box_1, \ldots, \Box_n\})$. Given terms $s_1, \ldots, s_n$, $C[s_1, \ldots, s_n]$ denotes the term obtained by simultaneously replacing every occurrence of $\Box_k$ by s_k.

Nota. C is *linear* in $\Box_1, \ldots, \Box_n$ if each hole occurs at exactly one position of C, denoted p_k.

Lemma (*critical pairs joinable implies local confluence*). Let R be a TRS (no termination assumed) such that every critical pair of R is joinable. Then R is locally confluent.

Proof. We show every local peak $b\ {}_R\!\leftarrow a \to_R c$ is joinable.

Let $A = \{a \mid \exists b, c : b \leftarrow_R a \to_R c,\ \nexists d, b \to_R^* d \leftarrow_R^* c\}$. Suppose $A \neq \emptyset$; since the strict subterm ordering is well-founded (definition of strict subterm order, Section 2), fix $a \in A$ minimal for the strict subterm relation among the elements of A, and fix b, c as above via rules

$l_1 \to r_1, l_2 \to r_2$, positions $p, q \in \mathrm{Pos}(a)$, substitutions σ_1, σ_2, with $a|_p = l_1\sigma_1$, $a|_q = l_2\sigma_2$, $b = a[r_1\sigma_1]_p$, $c = a[r_2\sigma_2]_q$.

We study the relative positions of p and q: we say $p \leq q$ if p is a prefix of q, i.e. $q = p \cdot q'$ (possibly $q' = \varepsilon$, giving $p = q$); and p, q are disjoint if neither is a prefix of the other. Exactly one of $p \leq q$, $q \leq p$, or p, q disjoint holds.

Case 1 (disjoint positions). Replacing a at p does not modify the subterm at q: $(a[r_1\sigma_1]_p)|_q = a|_q = l_2\sigma_2$, so the redex at q survives, giving $a[r_1\sigma_1]_p \to_R a[r_1\sigma_1]_p[r_2\sigma_2]_q$. Symmetrically, $a[r_2\sigma_2]_q \to_R a[r_2\sigma_2]_q[r_1\sigma_1]_p$. Since p, q are disjoint, the two replacements act on independent subtrees and commute: $a[r_1\sigma_1]_p[r_2\sigma_2]_q = a[r_2\sigma_2]_q[r_1\sigma_1]_p =: d$. Then $b \to_R d$ and $c \to_R d$: the peak is joinable.

Case 2 (p and q comparable). Without loss of generality, $p \leq q$ (the case $q \leq p$ is symmetric, exchanging the roles of the two rules). Write $q = p \cdot q'$ with $q' \in \mathrm{Pos}(l_1\sigma_1)$.

Case 2.1 (q' not below a variable of l_1). Then $q' \in \mathrm{Pos}(l_1)$ and $(l_1\sigma_1)|_{q'} = (l_1|_{q'})\sigma_1$.

If $q' = \varepsilon$: $l_1\sigma_1 = l_2\sigma_2$, the two rules apply to the same redex. If the two rules are identical up to a renaming of variables (bijective ρ with $l_2 = l_1\rho$, $r_2 = r_1\rho$), then, since $l_1\sigma_1 = l_1(\rho\sigma_2)$ forces σ_1 and $\rho\sigma_2$ to agree on $\mathcal{V}\mathrm{ar}(l_1)$, and $\mathcal{V}\mathrm{ar}(r_1) \subseteq \mathcal{V}\mathrm{ar}(l_1)$ (definition of TRS), this agreement extends to $r_1\sigma_1 = r_1(\rho\sigma_2) = r_2\sigma_2$, so $b = c$, trivially joinable. Otherwise, unifying l_1, l_2 yields a most general unifier μ and critical pair $(r_1\mu, r_2\mu)$; by hypothesis joinable, giving d with $r_1\mu \to_R^* d$, $r_2\mu \to_R^* d$. There exists θ with $\sigma_1 = \mu\theta$ on $\mathcal{V}\mathrm{ar}(l_1)$, $\sigma_2 = \mu\theta$ on $\mathcal{V}\mathrm{ar}(l_2)$; by stability under substitution and context, $b \to_R^* a[d\theta]_p$ and $c \to_R^* a[d\theta]_p$.

If $q' \neq \varepsilon$: q' is a non-variable position of l_1, so unifying $l_1|_{q'}$ and l_2 yields a most general unifier μ and critical pair $(r_1\mu,\ l_1[r_2]_{q'}\mu)$; by hypothesis joinable in some d. As above, via θ, $b \to_R^* a[d\theta]_p$ and $c \to_R^* a[d\theta]_p$.

Case 2.2 (q' below a variable of l_1). There exist $x \in \mathcal{V}\mathrm{ar}(l_1)$, $p_x \in \mathrm{Pos}(l_1)$ with $l_1|_{p_x} = x$, and $q'' \in \mathrm{Pos}(x\sigma_1)$ with $q' = p_x \cdot q''$, $(x\sigma_1)|_{q''} = l_2\sigma_2$. Define $\sigma_1'(x) = (x\sigma_1)[r_2\sigma_2]_{q''}$, $\sigma_1'(y) = \sigma_1(y)$ for $y \neq x$. Then $c = a[r_2\sigma_2]_q = a[l_1\sigma_1']_p \to_R a[r_1\sigma_1']_p =: w$. Since $x\sigma_1 \to_R x\sigma_1'$ is precisely the step $l_2\sigma_2 \to r_2\sigma_2$, rewriting each of the (possibly $k \geq 0$) occurrences of $x\sigma_1$ inside $r_1\sigma_1$ gives $r_1\sigma_1 \to_R^* r_1\sigma_1'$, hence, by stability under context, $b = a[r_1\sigma_1]_p \to_R^* a[r_1\sigma_1']_p = w$: the peak is joinable.

All local peaks are joinable: R is locally confluent. $\square$

> **Theorem** (*Critical Pair Theorem (Knuth–Bendix)*). A terminating TRS is confluent if and only if both terms of each of its critical pairs reduce to a common term.

Proof. Let R be a terminating TRS.

(⇒) Assume R is confluent. Let (s, s') be a critical pair, arising from rules $l_1 \to r_1, l_2 \to r_2$, position p, and most general unifier σ, with $s = (l_1\sigma)[r_2\sigma]_p$ and $s' = r_1\sigma$:

$$
\begin{tikzcd}
 & l_1\sigma \arrow[dl, "l_2 \to r_2"'] \arrow[dr, "l_1 \to r_1"] & \\
s = (l_1\sigma)[r_2\sigma]_p & & s' = r_1\sigma
\end{tikzcd}
$$

Since $l_1\sigma \to_R s$ and $l_1\sigma \to_R s'$, confluence gives a term u with $s' \to_R^* u \,{}^*\!\!\leftarrow_R s$: every critical pair is joinable.

(⇐) Assume every critical pair is joinable. By the lemma on critical pairs joinable implies local confluence, R is locally confluent. Since R terminates, Newman's Lemma gives that R is confluent. □

Theorem (*Non-overlap implies local confluence (Dershowitz)*). Let R be a TRS such that no left-hand side of a rule of R unifies with a non-variable subterm of any left-hand side of R (of the same or of another rule, variables of distinct rules or of distinct instances of the same rule always being treated as disjoint), except for the trivial self-overlap of a rule with itself at the root. Then R is locally confluent, with no termination assumption whatsoever.

Proof. By hypothesis, R has no non-trivial overlap (definition of overlap): for any two rules $l_1 \to r_1, l_2 \to r_2$ (possibly the same rule, renamed apart), no non-variable position $p \in \mathrm{Pos}(l_1)$ and substitution σ satisfy $l_1|_p\sigma = l_2\sigma$, except in the excluded trivial case ($l_1 = l_2$, $p = \varepsilon$). By the definition of critical pair, which is built precisely from such an overlap at a non-variable position $p \neq \varepsilon$ of l_1 (the trivial root self-overlap being excluded by convention, as it always yields the trivial pair (r_1, r_1)), it follows that R has no critical pair at all: the set of critical pairs of R is empty. Vacuously, every critical pair of R is joinable (there is none to fail joinability). By the lemma on critical pairs joinable implies local confluence (which assumes no termination), R is locally confluent. □

Remark. This theorem isolates exactly what non-overlap alone buys us: local confluence, unconditionally, with no need for termination and no need for left-linearity. It does *not* by itself give global confluence for a non-terminating system: Newman's Lemma, the classical route from local to global confluence, needs termination, which is not assumed here. The *additional requirement* needed to bypass Newman's Lemma entirely and reach global confluence directly, even without termination, is exactly left-linearity: turning non-overlap into full orthogonality (definition of Orthogonal system, Section 7.5) and this theorem's conclusion (local confluence) into the strictly stronger conclusion of the theorem on orthogonal confluence (Rosen): global confluence. The present theorem is thus best read as isolating the precise contribution of each hypothesis: non-overlap alone secures local confluence; left-linearity is what is additionally needed to lift this, unconditionally, all the way to global confluence.

6 Completion

Not every TRS is confluent, and not every equational theory admits an obviously confluent orientation. The *Knuth–Bendix completion procedure* attempts to construct, from an arbitrary equational presentation, a convergent TRS defining the same equational theory, by repeatedly detecting and resolving the failures of local confluence exhibited by critical pairs (Section 5), whose computation in turn rests on the unification algorithm of Section 4.

Definition (*Equivalent TRS*). Two TRS R, S are *equivalent* if $\to_R^* = \to_S^*$.

Definition (*Reduction order*). A *reduction order* on $\mathcal{T}(\Sigma, \mathcal{X})$ is a strict order $\succ$ that is (i) well-founded, (ii) stable under substitution ($s \succ t \Rightarrow s\sigma \succ t\sigma$), (iii) stable under context

$(s \succ t \Rightarrow C[s] \succ C[t])$. A rule $l \to r$ is *compatible* with $\succ$ if $l \succ r$; a TRS R is compatible with $\succ$ if every rule is.

Lemma (*compatible TRS terminate*). If R is compatible with a reduction order $\succ$, then R terminates.

Proof. An infinite reduction $t_0 \to t_1 \to \cdots$ would give, by stability under substitution and context at each redex, $t_0 \succ t_1 \succ \cdots$, an infinite $\succ$-decreasing sequence, contradicting well-foundedness. □

Definition (*Configuration transition relation*). Fix a finite E_0 and a reduction order $\succ$. A *configuration* is a pair (E, R), E a set of equations, R a TRS. As with the unification transition relation of Section 4 (definition of unification transition relation), we write $(E, R) \vdash (E', R')$ to mean (E', R') is obtained from (E, R) by one application of one of the six inference rules below, and $\vdash^*$ for its reflexive-transitive closure.

Definition (*Knuth–Bendix completion*). The completion procedure is the transition system on configurations (E, R), starting from $(E_0, \emptyset)$, governed by the following rules.

Orient. $(E \cup \{s \approx t\}, R) \vdash (E, R \cup \{s \to t\})$ if $s \succ t$ (symmetrically for $t \succ s$).

Delete. $(E \cup \{s \approx s\}, R) \vdash (E, R)$.

Simplify. $(E \cup \{s \approx t\}, R) \vdash (E \cup \{s' \approx t\}, R)$ if $s \to_R s'$.

Compose. $(E, R \cup \{l \to r\}) \vdash (E, R \cup \{l \to r'\})$ if $r \to_R r'$.

Collapse. $(E, R \cup \{l \to r\}) \vdash (E \cup \{l' \approx r\}, R)$ if $l \to_R l'$ using another rule of R at a non-root, non-variable position of l.

Deduce. $(E, R) \vdash (E \cup \{s \approx t\}, R)$ if (s, t) is a critical pair of R.

A run $(E_0, \emptyset) \vdash (E_1, R_1) \vdash \cdots$ *succeeds* if it reaches $(\emptyset, R)$: R is then the *completion* of E_0 w.r.t. $\succ$.

Definition (*fair run*). A successful run $(E_0, \emptyset) \vdash (E_1, R_1) \vdash \cdots \vdash (E_N, R_N) = (\emptyset, R)$ is *fair* if, for every two rules $l_1 \to r_1$, $l_2 \to r_2 \in R$ present in the final TRS R, every critical pair of $\{l_1 \to r_1,\ l_2 \to r_2\}$ (computed from these rules *in their final form*) was added to some E_i by an application of Deduce at a step i where both rules were already present in R_i in that same final form.

Nota. Fairness is not automatic from mere success: a run may reach $(\emptyset, R)$ by emptying E before ever invoking Deduce on some critical pair of the rules of the final R. For instance, from $E_0 = \{a \approx b,\ a \approx c\}$ (a, b, c distinct constants, $a \succ b$, $a \succ c$), orienting both equations gives $R = \{a \to b,\ a \to c\}$ and an immediately empty E: the run succeeds, yet the critical pair (b, c) arising from these two rules (sharing the left-hand side a; this is *not* the excluded trivial self-overlap of a rule with itself, since the two rules are distinct) was never Deduced. Indeed R is not locally confluent here (a reduces to both b and c, with no further rule relating them). Ensuring fairness in practice requires a scheduling strategy for the non-deterministic transition system that never indefinitely postpones an available Deduce step; we take this as a standing hypothesis in the theorem below, as is standard in the literature on completion.

Definition (*equational closure of a configuration*). For (E, R), let $=_{E,R}$ be the smallest congruence, stable under substitution, containing $\to_R$ and every instance $s\sigma \approx t\sigma$ for $s \approx t \in E$.

Lemma (*soundness of completion steps*). If $(E, R) \vdash (E', R')$, then $=_{E,R} = =_{E',R'}$.

Proof. Orient: the generator $s\sigma \approx t\sigma$ is reclassified as $s\sigma \to t\sigma$; unchanged as a relation. *Delete:* $(s\sigma, s\sigma)$ is already given by reflexivity. *Simplify:* since $s \to_R s'$ holds in both configurations, $s\sigma \approx t\sigma$ and $s'\sigma \approx t\sigma$ are interderivable via $s\sigma \to_R s'\sigma \approx t\sigma$. *Compose:* similarly, using the shared reduction $r\sigma \to_R^* r'\sigma$. *Collapse:* using the shared $l\sigma \to_R^* l'\sigma \approx r\sigma$. *Deduce:* by construction, a critical pair (s, t) satisfies $s\ {}_R{\leftarrow}\ u \to_R t$ for a common ancestor u, so $s \approx t$ is already derivable from R alone. □

Theorem (*Correctness of Knuth–Bendix completion*). If a *fair* run $(E_0, \emptyset) \vdash^* (\emptyset, R)$ succeeds, then R is convergent and equivalent to E_0: $\to_R^* = =_{E_0}$.

Proof. Termination. We show, by induction on the length of the run, that R_i is compatible with $\succ$ at every configuration (E_i, R_i). This holds vacuously for $R_0 = \emptyset$. Orient only adds a rule $s \to t$ with $s \succ t$ by construction, preserving compatibility of the rest. Delete, Simplify, Deduce do not modify R_i. Compose replaces a rule $l \to r$ (compatible: $l \succ r$, by the induction hypothesis) by $l \to r'$ where $r \to_{R_i}^* r'$ using the (by induction hypothesis, compatible) remaining rules of R_i: by the lemma of compatible TRS terminate's underlying mechanism, $r \to_{R_i} r'$ gives $r \succ r'$ at each step, hence $r \succ r'$ overall by transitivity of $\succ$, and so $l \succ r \succ r'$ gives $l \succ r'$ (transitivity again): the new rule remains compatible. Collapse only removes a rule. Hence compatibility with $\succ$ is preserved at every step, so the final R is compatible with $\succ$, and by the lemma of compatible TRS terminate, R terminates.

Confluence. By soundness applied along the whole run, $=_{E_0,\emptyset} = =_{\emptyset,R}$. By fairness, every critical pair of the final R (computed from the rules of R in their final form) was added by Deduce to some E_i at a step where both contributing rules were already present in their final form; since the run ends with $E = \emptyset$, that pending equation was necessarily later consumed (by Orient, Delete, or Simplify) without altering $=_{E,R}$ (soundness), so every such critical pair (s, t) satisfies $s \leftrightarrow_R^* t$. Since R terminates, Newman's Lemma converts this convertibility into joinability once local confluence is known; by the Critical Pair Theorem, joinability of every critical pair gives local confluence of R, hence, with termination, confluence by Newman's Lemma.

Equivalence. By soundness, $=_{E_0} = =_{E_0,\emptyset} = =_{\emptyset,R} = \leftrightarrow_R^*$. Since R is confluent, $\to_R^* = =_{E_0}$. □

Remark. Limits of the procedure. A run of completion need not succeed: at any Deduce step the resulting equation may fail to be orientable by the fixed order $\succ$ (a *failure* of completion, requiring a different or more powerful order), or the procedure may produce infinitely many rules without ever exhausting E (*divergence*, distinct from non-termination of the TRS produced). A third, more insidious limit is *unfairness*: even a run that does succeed, in the bare sense of reaching $(\emptyset, R)$, need not be fair (definition of fair run above), in which case the resulting R need not be confluent at all, as the counterexample above shows. This is not a defect of our presentation: it reflects the genuine limits of the method, and is one of the motivations for variants such as unfailing completion, which we do not develop here.

7 Modularity

We now study the extent to which the properties introduced above are preserved under *disjoint unions* of term rewriting systems: given two systems $R_1 = (\Sigma_1, \mathcal{R}_1)$ and $R_2 =$

$(\Sigma_2, \mathcal{R}_2)$ with $\Sigma_1 \cap \Sigma_2 = \emptyset$, each individually enjoying a property P, does $R_1 \cup R_2$ also enjoy P? We show that the answer is a resounding *yes* for confluence (the theorem on the union of confluent TRS), and an equally resounding *no* for termination in general (the proposition on non-modularity of termination), unless left-linearity is additionally assumed (the theorem on the union of convergent left-linear TRS). We also treat, as a structurally related but independent question, the confluence of *orthogonal* systems (Section 7.5).

> **Theorem** (*Union of confluent TRS*). The union of two confluent TRS is confluent if they don't share any function symbol (or constant).

Note. The advantage of having distinguished signatures is that in both systems no rule of one system rewrites a term into the other system's territory in an uncontrolled way. On the other hand, the union means that we can have *mixed* terms, combining symbols of both signatures. The whole difficulty of this theorem lies in analysing how rewriting acts on such mixed terms.

7.1 Aliens, skeletons, and rank

> **Definition** (*pure term*). Let $R_i = (\Sigma_i, \mathcal{R}_i)$. A term t is *pure* for R_i if all of its symbols belong to Σ_i: $t \in T(\Sigma_i, \mathcal{X})$.

> **Definition** (*Alien term*). Let $t \in \mathcal{T}(\Sigma, \mathcal{X})$ be a term whose root belongs to Σ_i (we say that t is *i-rooted*: $\mathrm{root}(t) \in \Sigma_i$). A subterm $s = t|_p$ (for some $p \in \mathrm{Pos}(t)$, $p \neq \varepsilon$) is called an *alien* of t (for R_i) if $\mathrm{root}(s) \in \Sigma_j$, $j \neq i$, while every proper prefix of p carries a symbol of Σ_i. Equivalently: s is a maximal subterm of t whose root does not belong to Σ_i.
> No condition is imposed on the symbols occurring strictly below the root of s: an alien may itself contain further aliens (rooted back in Σ_i, or beyond).

Example. Let $\Sigma_1 = \{f, a, b\}$, $\Sigma_2 = \{g\}$, and $t = f(g(f(a), y), b)$. Then t is 1-rooted, and $g(f(a), y)$ is an alien of t (for R_1), even though it itself contains $f(a)$, which is rooted back in Σ_1. This illustrates why no purity condition is imposed on the interior of an alien.

> **Definition** (*term preserver*). Let $R_1 = (\Sigma_1, \mathcal{R}_1)$ and $R_2 = (\Sigma_2, \mathcal{R}_2)$ be two rewrite systems with disjoint signatures. A rewrite rule $\ell \to r \in \mathcal{R}_1$ is a *term preserver* if $\mathcal{F}(r) \subseteq \Sigma_1 \cup \mathcal{V}\mathrm{ar}(\ell)$ (it introduces no new symbol from Σ_2). Concretely, for a step $t \xrightarrow[p]{\ell \to r}_{R_1} t'$: any alien of t outside p is unchanged in t', and any alien inside $t|_p$ is only substituted for a variable of ℓ, hence reappears unmodified (possibly copied or deleted, never altered). R_1 may rearrange, duplicate, or erase aliens from R_2, but never fabricate a new one.

> **Definition** (*collapsing rule*). A rule $l \to_R r \in \mathcal{R}_i$ is *collapsing* if $r \in \mathcal{X}$. A rewrite step is *collapsing* if it uses a collapsing rule.

Nota. A term t is *preserved* (for $R_1 \cup R_2$) if no reduction sequence issued from t, finite or infinite, contains a collapsing step.

Definition (*skeleton decomposition*). Let t be i-rooted with aliens $s_1, \dots, s_n$ at (pairwise disjoint) positions $p_1, \dots, p_n$. There exists a unique linear context $C \in \mathcal{T}(\Sigma_i, \mathcal{X} \cup \{\Box_1, \dots, \Box_n\})$ with holes at exactly $p_1, \dots, p_n$ such that $t = C[s_1, \dots, s_n]$. We call C the *skeleton* of t.

7.2 The mixed local peak

Lemma (*joinability of mixed local peaks*). Let $R_1 = (\Sigma_1, \mathcal{R}_1)$ and $R_2 = (\Sigma_2, \mathcal{R}_2)$ be two TRS with $\Sigma_1 \cap \Sigma_2 = \emptyset$ (no left-linearity assumed). Then $R_1 \cup R_2$ satisfies: for every local peak $t \ {}_{R_1}\!\leftarrow s \rightarrow_{R_2} u$, there exists w such that $t \rightarrow^*_{R_2} w$ and $u \rightarrow^*_{R_1 \cup R_2} w$ (the peak is joinable in the union; a pure R_1-path for u need not exist when the rule of R_1 involved is not left-linear, see the proof of Case $p < q$ below).

$$\begin{tikzcd}
 & t \arrow[dr, dashed, "{R_2,\ *}"] & \\
s \arrow[ur, "R_1"] \arrow[dr, "R_2"'] & & w \\
 & u \arrow[ur, dashed, "{R_1 \cup R_2,\ *}"'] &
\end{tikzcd}$$

Proof. By definition of a rewrite step, there exist rules $l_1 \rightarrow_R r_1 \in \mathcal{R}_1$, $l_2 \rightarrow_R r_2 \in \mathcal{R}_2$, positions $p, q \in \mathrm{Pos}(s)$, substitutions σ_1, σ_2 with $s|_p = l_1\sigma_1$, $s|_q = l_2\sigma_2$, $t = s[r_1\sigma_1]_p$, $u = s[r_2\sigma_2]_q$.

Preliminary claim. Let $i \in \{1, 2\}$, $l \rightarrow_R r \in \mathcal{R}_i$, σ a substitution, $q' \in \mathrm{Pos}(l\sigma)$. Exactly one of the following holds: (a) q' is not below a variable of l, in which case $q' \in \mathrm{Pos}(l)$ and $\mathrm{root}((l\sigma)|_{q'}) = \mathrm{root}(l|_{q'}) \in \Sigma_i$; (b) q' is below a variable of l: there exist $x \in \mathcal{V}\mathrm{ar}(l)$, $p_x \in \mathrm{Pos}(l)$ with $l|_{p_x} = x$, and $q'' \in \mathrm{Pos}(x\sigma)$, such that $q' = p_x \cdot q''$, $(l\sigma)|_{q'} = (x\sigma)|_{q''}$. This follows by a straightforward induction on q', using that $l \notin \mathcal{X}$ and $l \in T(\Sigma_i, \mathcal{X})$.

The case $p = q$ cannot occur. If $p = q$, then $s|_p = l_1\sigma_1 = l_2\sigma_2$. Applying the preliminary claim to each side at $q' = \varepsilon$ (always case (a), since ε has no proper prefix) gives $\mathrm{root}(l_1\sigma_1) = \mathrm{root}(l_1) \in \Sigma_1$ and $\mathrm{root}(l_2\sigma_2) = \mathrm{root}(l_2) \in \Sigma_2$, contradicting $\Sigma_1 \cap \Sigma_2 = \emptyset$.

Case $p \parallel q$ (disjoint). Replacing at p leaves the subterm at q unchanged: $(s[r_1\sigma_1]_p)|_q = s|_q = l_2\sigma_2$, so $s[r_1\sigma_1]_p \rightarrow_{R_2} s[r_1\sigma_1]_p[r_2\sigma_2]_q$. Symmetrically $s[r_2\sigma_2]_q \rightarrow_{R_1} s[r_2\sigma_2]_q[r_1\sigma_1]_p$. The two replacements act on disjoint subtrees, hence commute: let w be the common term; $t \rightarrow_{R_2} w$, $u \rightarrow_{R_1} w$ (in particular $u \rightarrow^*_{R_1 \cup R_2} w$).

Case $p < q$. Write $q = p \cdot q'$, $q' \in \mathrm{Pos}(l_1\sigma_1)$. By the preliminary claim, case (a) would give $\mathrm{root}((l_1\sigma_1)|_{q'}) \in \Sigma_1$, contradicting $(l_1\sigma_1)|_{q'} = s|_q = l_2\sigma_2$ (rooted in Σ_2). So case (b) holds: $q' = p_x \cdot q''$, $l_1|_{p_x} = x$, $(x\sigma_1)|_{q''} = l_2\sigma_2$.

Let $k \geq 1$ be the number of occurrences of x in l_1, at positions $p_x^{(1)}, \dots, p_x^{(k)}$ (with $p_x^{(1)} := p_x$ matching the occurrence identified above). Since $l_1\sigma_1$ holds the identical value $x\sigma_1$ at each of these k positions (substitution is uniform), and $(x\sigma_1)|_{q''} = l_2\sigma_2$ exhibits an $\mathcal{R}_2$-redex *within this shared value*, the identical redex (rule $l_2 \rightarrow r_2$, substitution σ_2, relative position q'') is present at each of the k corresponding absolute positions $p \cdot p_x^{(i)} \cdot q''$ of s. Rewriting them one at a time (order immaterial, being pairwise disjoint occurrences) gives a chain of k R_2-steps

$$s = s^{(0)} \to_{R_2} s^{(1)} \to_{R_2} \cdots \to_{R_2} s^{(k)} =: u',$$

where $s^{(1)} = s[r_2\sigma_2]_{p \cdot p_x^{(1)} \cdot q''} = s[r_2\sigma_2]_q = u$ (the step given by hypothesis) and each subsequent step updates the next occurrence. After all k steps, every occurrence of x in $u'|_p$ holds the updated value $\sigma_1'(x) := (x\sigma_1)[r_2\sigma_2]_{q''}$ uniformly (defining $\sigma_1'(y) := \sigma_1(y)$ for $y \neq x$), so $u'|_p = l_1\sigma_1'$ matches l_1 via σ_1': apply $l_1 \to r_1$ at p,

$$u' = s[l_1\sigma_1']_p \to_{R_1} s[r_1\sigma_1']_p =: w.$$

Hence $u = s^{(1)} \to_{R_2}^{k-1} u' \to_{R_1} w$, i.e. $u \to_{R_1 \cup R_2}^* w$: in general this mixes R_2 and R_1 steps, and need not be a pure R_1-path when $k > 1$ (i.e. when l_1 is not left-linear in x); if l_1 happens to be linear in x ($k = 1$), $u = u'$ and the path is the single R_1-step $u \to_{R_1} w$, recovering the pure-R_1 conclusion in that special case.

On the other side, since $x\sigma_1 \to_{R_2} x\sigma_1'$ is precisely the step $l_2\sigma_2 \to r_2\sigma_2$, rewriting each of the (possibly $k' \geq 0$) occurrences of $x\sigma_1$ inside $r_1\sigma_1$ gives $r_1\sigma_1 \to_R^* r_1\sigma_1'$, hence, by stability under context, $t = s[r_1\sigma_1]_p \to_R^* s[r_1\sigma_1']_p = w$ (this holds regardless of k', in particular $k' = 0$ if $x \notin \mathcal{V}\mathrm{ar}(r_1)$).

Case $q < p$ is symmetric to the previous case, exchanging the roles of the two rules: one obtains $u \to_{R_1} w$ (pure R_1-step, since $u = s[r_2\sigma_2]_q$ is unaffected by any multiplicity issue on the R_1 side of the argument) and $t \to_{R_1 \cup R_2}^* w$, mixing R_1 then R_2 steps in general (with a pure R_2-path recovered when the $\mathcal{R}_2$-rule involved is left-linear in the relevant variable).

Since the cases exhaust all possibilities, every local peak $t \;{}_{R_1}\!\leftarrow s \to_{R_2} u$ is joinable, with $t \to_{R_2}^* w$ always achieved by a pure R_2-path, and $u \to_{R_1 \cup R_2}^* w$ in general (pure R_1 only under a left-linearity restriction on the rules involved). ☐

Lemma (*extension of confluence to extra variables*). Let $R_i = (\Sigma_i, \mathcal{R}_i)$ be confluent on $\mathcal{T}(\Sigma_i, \mathcal{X})$, and let $\mathcal{X}' \supseteq \mathcal{X}$. Then R_i is confluent on $\mathcal{T}(\Sigma_i, \mathcal{X}')$.

Proof. Let $C \to_{R_i}^* C_1$, $C \to_{R_i}^* C_2$ in $\mathcal{T}(\Sigma_i, \mathcal{X}')$. Let $\Box_1, \ldots, \Box_n$ be the variables of $\mathcal{X}' \setminus \mathcal{X}$ occurring in C (finitely many, since C is a finite term). Since $\mathcal{V}\mathrm{ar}(r) \subseteq \mathcal{V}\mathrm{ar}(l)$ for every rule of R_i (definition of TRS), no rewrite step can introduce a variable not already present in the term being rewritten: hence $C_1, C_2 \in \mathcal{T}(\Sigma_i, \mathcal{X} \cup \{\Box_1, \ldots, \Box_n\})$ as well, the same finite extended set. Since every rule's left-hand side is non-variable, no rule matches a redex rooted at a $\Box_k$; rewriting propagates the $\Box_k$'s exactly as ordinary variables of $\mathcal{X}$. Renaming $\Box_1, \ldots, \Box_n$ to fresh variables of $\mathcal{X}$, not occurring elsewhere in C, C_1, C_2, via a bijection ρ, yields (by stability under substitution, applied along the whole reduction sequences) $C^\rho \to_{R_i}^* C_1^\rho$, $C^\rho \to_{R_i}^* C_2^\rho$ in $\mathcal{T}(\Sigma_i, \mathcal{X})$. By confluence there, D with $C_1^\rho \to_{R_i}^* D$, $C_2^\rho \to_{R_i}^* D$; applying ρ^{-1}, $C_1 \to_{R_i}^* D^{\rho^{-1}}$, $C_2 \to_{R_i}^* D^{\rho^{-1}}$. ☐

Lemma (*skeleton-alien commutation*). Let $t = C[s_1, \ldots, s_n]$ be i-rooted, and let $t \xrightarrow[p]{l \to r} t'$ be a single step. Exactly one holds: (a) $p_k \leq p$ for a (necessarily unique) k, i.e. p lies at or strictly below the alien position p_k: then $s_k \to_{R_1 \cup R_2} s_k'$ for the specific rule $l \to r$ matching $t|_p$ (which may belong to either $\mathcal{R}_i$ or $\mathcal{R}_j$ when $p_k < p$ strictly, since s_k need not itself be pure and may contain further alternations; but necessarily $l \to r \in \mathcal{R}_j$ when $p = p_k$, since s_k's own root lies in Σ_j), $t' = C[s_1, \ldots, s_k', \ldots, s_n]$, and this step is non-collapsing whenever t is preserved. (b) p is disjoint from every p_j, or p is a strict ancestor of some p_j ($p < p_j$), i.e. p genuinely lies within the skeleton, never reaching or passing a hole: then $l \to r \in \mathcal{R}_i$, and $C \to_{R_i} C'$ (a skeleton step, $\Box_k$'s treated as opaque constants during matching), $t' = C'[s_1, \ldots, s_n]$, provided t is preserved; without this proviso, a collapsing rule of $\mathcal{R}_i$ matching a subtree of C that happens to contain one or more holes could, in principle, produce a t' no longer of this clean form (exposing part of one or more aliens directly); this degenerate possibility is precisely excluded by preservedness, and is otherwise handled separately, at a coarser grain, by the lemma of first-switch decomposition.

Proof. Since $p_1, \ldots, p_n$ are pairwise disjoint and p is a single position, exactly one of the following holds for p relative to the family $p_1, \ldots, p_n$: $p_k \leq p$ for some (necessarily unique, by pairwise disjointness) k; or p is disjoint from every p_j; or p is a strict ancestor of some p_j (possibly several, if they share p as a common ancestor). This exhausts all cases, giving exactly (a) or (b) as stated.

Case (a), sub-case $p = p_k$. Then $t|_p = s_k$ entirely, matched by $l\sigma = s_k$. Since substitution preserves the root of a non-variable term, $\mathrm{root}(l) = \mathrm{root}(s_k) \in \Sigma_j$ (definition of alien term), forcing $l \to r \in \mathcal{R}_j$. If $l \to r$ is non-collapsing, $r\sigma$ is again rooted in Σ_j (a non-collapsing rule of $\mathcal{R}_j$ has $r \in T(\Sigma_j, \mathcal{X})$), so $s'_k := r\sigma$ is a new alien of index j, and $t' = t[r\sigma]_{p_k} = C[s_1, \ldots, s'_k, \ldots, s_n]$: this matches the general pattern of case (a) with $p' = \varepsilon$. If preserved, this step (being non-collapsing) does not affect preservedness. If $l \to r$ is collapsing, $r = y \in \mathcal{X}$ and $t' = t[y\sigma]_{p_k}$ exposes (a subterm of, or the whole of) s_k at position p_k: this is a collapsing step of t's own reduction, hence excluded when t is preserved.

Case (a), sub-case $p_k < p$ strictly. Then $t|_p = s_k|_{p'}$ for $p = p_k \cdot p'$, $p' \neq \varepsilon$. Since s_k need not be pure (it may itself contain aliens, i.e. further alternations back into Σ_i or beyond, per the definition of alien term with no purity condition on its interior), $\mathrm{root}(t|_p)$ may lie in Σ_i or Σ_j: correspondingly $l \to r$ is some rule of $\mathcal{R}_i \cup \mathcal{R}_j = \mathcal{R}_1 \cup \mathcal{R}_2$ matching $t|_p$, and by stability of replacement under nested positions,

$$t' = t[r\sigma]_p = C[s_1, \ldots, s_k[r\sigma]_{p'}, \ldots, s_n] = C[s_1, \ldots, s'_k, \ldots, s_n],$$

where $s'_k = s_k[r\sigma]_{p'}$, i.e. $s_k \to_{R_1 \cup R_2} s'_k$ using the same rule and position p'. If t is preserved, every step of every reduction issued from t is non-collapsing; since $s_k \to s'_k$ is (the image, under stability under context, of) such a step, it is non-collapsing.

Case (b). Since p is disjoint from, or a strict ancestor of, every p_j it is comparable to, $t|_p$ never reaches or passes a hole: it is matched entirely within C (the $\Box_k$'s acting as opaque constants, since σ only instantiates variables of $\mathcal{X}$, and $l \in \mathcal{T}(\Sigma_i, \mathcal{X})$ does not mention the $\Box_k$'s). Since $t|_p = C|_p$ and C is pure for Σ_i, $\mathrm{root}(t|_p) \in \Sigma_i$, so $l \to r \in \mathcal{R}_i$. Assume t is preserved (so this step, being part of a reduction issued from t, is non-collapsing). Then the step is exactly a step of C (as a term over the extended variable set, with holes as variables): $C \to_{R_i} C'$, and $t' = C'[s_1, \ldots, s_n]$, the aliens propagating (duplicated or erased according to the multiplicities of the $\Box_k$'s in r) exactly as ordinary variables of $\mathcal{X}$ would.

(Without assuming t preserved, a collapsing rule of $\mathcal{R}_i$ could match a subterm of C containing one or more holes and collapse to a variable bound to a subterm containing those holes' content, yielding a t' that mixes skeleton and alien material in a way not of the clean form $C'[s_1, \ldots, s_n]$; this scenario is exactly excluded here by preservedness, and is the mechanism tracked, at the level of an entire reduction rather than a single step, by the case (S) of the lemma of first-switch decomposition.) $\Box$

Lemma (*purity is preserved under own-system rewriting*). Let $s \in \mathcal{T}(\Sigma_j, \mathcal{X})$ be pure for R_j, and $s \to_{R_j} s'$. Then $s' \in \mathcal{T}(\Sigma_j, \mathcal{X})$.

Proof. Write $s \xrightarrow[q]{\ell \to r} s'$ with $\ell, r \in \mathcal{T}(\Sigma_j, \mathcal{X})$. Every symbol of r is in Σ_j; every subterm of s substituted for a variable of ℓ is itself pure (a subterm of a pure term is pure). Hence $r\sigma$, and thus s', is pure. $\square$

Lemma (*subterms of preserved terms are preserved*). Let t be preserved and $s \sqsubset t$. Then s is preserved.

Proof. Suppose a reduction $s = s_0 \to s_1 \to \cdots$ contains a collapsing step $s_k \to s_{k+1}$. Writing $t = C[s]$, stability under context reinjects each step as $C[s_l] \to C[s_{l+1}]$, using the same rule at the same relative position, hence the same collapsing status, producing a reduction from t containing a collapsing step: contradiction. $\square$

7.3 Rank and the induction measure

Nota. We reuse the strict subterm order $\sqsubset$ and its well-foundedness (definition of strict subterm order and the lemma of well-foundedness of the strict subterm order, both introduced in Section 2) as the induction measure below.

Definition (*rank*). We define $\text{rank}(t) \in \mathbb{N}^*$ by well-founded recursion on $\sqsubset$:

$$\text{rank}(t) = \begin{cases} 1 & \text{if } t \text{ is pure (for } R_1 \text{ or } R_2\text{)}, \\ 1 + \max\{\text{rank}(s) : s \text{ alien of } t\} & \text{otherwise.} \end{cases}$$

Lemma (*strict rank decrease for aliens*). If s is an alien of t, then $\text{rank}(s) < \text{rank}(t)$.

Proof. Since t is not pure (it has an alien), $\text{rank}(t) = 1 + \max\{\text{rank}(s') : s' \text{ alien of } t\} \geq 1 + \text{rank}(s) > \text{rank}(s)$. $\square$

Example. Let $\Sigma_1 = \{f, a, b\}$, $\Sigma_2 = \{g\}$, $t = f(g(f(a), y), b)$ as before. The alien of t is $s_1 = g(f(a), y)$, which is not pure (it contains $f(a)$, rooted in Σ_1): its own alien is $f(a)$, which is pure. Hence $\text{rank}(f(a)) = 1$, $\text{rank}(g(f(a), y)) = 2$, $\text{rank}(t) = 3$. This illustrates why an unbounded number of alternations between the two signatures is possible, and why rank (not merely "pure vs. not pure") is the correct induction measure.

Lemma (*reduction decomposition*). Let $t = C_0[s_1, \ldots, s_n]$ be preserved, i-rooted. Let $t = t_0 \to t_1 \to \cdots \to t_m$ be any finite reduction. Then there exist, for each k, a skeleton C_k and a list of aliens $(u_1^{(k)}, \ldots)$ with $t_k = C_k[u_1^{(k)}, \ldots]$, and for each $k < m$ exactly one of: (a) $C_{k+1} = C_k$ and a unique alien is rewritten internally; (b) $C_k \to_{R_i} C_{k+1}$ (a skeleton step), with the alien list updated by duplication/erasure. Consequently, discarding type-(a) steps yields $C_0 \to_{R_i}^* C_m$.

Proof. By induction on m. The base case $k = 0$ is given: $C_0, (s_1, \ldots, s_n)$ is the skeleton decomposition of $t = t_0$ by hypothesis.

For the inductive step, assume $t_k = C_k[u_1^{(k)}, \ldots, u_{n_k}^{(k)}]$ is the (unique, by definition of skeleton decomposition) skeleton decomposition of t_k, and apply Lemma [skeleton-alien commutation] to t_k, viewed with this skeleton and alien list: the exceptional collapsing sub-case of (b) cannot occur since t is preserved (hence so is t_k, being reached from t along a reduction issued from t), so every step of the reduction falls under case (a) or (b) as stated.

In case (a), the rewritten alien $u_l^{(k+1)}$ still has its root in the same foreign signature as $u_l^{(k)}$: either the whole alien is rewritten at its own root ($p = p_k$ in the notation of Lemma [skeleton-alien commutation]), necessarily via a rule of that alien's own system, whose right-hand side (being non-collapsing, as just argued) keeps the same root signature; or the rewrite occurs strictly inside the alien, leaving its root untouched. Either way, $u_l^{(k+1)}$ remains an alien at the same position p_l of C_k, so $C_{k+1} = C_k, (u_1^{(k+1)}, \ldots)$ is again the skeleton decomposition of t_{k+1}, licensing the next application of the lemma. In case (b), the aliens themselves are untouched as terms (only duplicated, erased, or repositioned by the skeleton-level rule), so they equally remain valid aliens of t_{k+1} at their (possibly new) positions in C_{k+1}. This closes the induction. ☐

Lemma (*Confluence of preserved terms*). Let R_1, R_2 be confluent TRS with $\Sigma_1 \cap \Sigma_2 = \emptyset$. Every preserved term t is confluent for $R_1 \cup R_2$.

Proof. Let $R = R_1 \cup R_2$ and $T = \{t \text{ preserved} \mid \exists t_1, t_2 : t_1 \ {}^{+}\!\leftarrow_R t \to_R^+ t_2,\ \nexists w,\ t_1 \to_R^* w \ {}^{*}\!\leftarrow_R t_2\}$. Suppose $T \neq \emptyset$; let $n_0 = \min\{\operatorname{rank}(t) : t \in T\}$, fix $t \in T$ with $\operatorname{rank}(t) = n_0$ and witnesses t_1, t_2.

Case $n_0 = 1$. t is pure, say for R_1. No rule of R_2 matches any position of t, so every R-step from t is an R_1-step, and by Lemma [purity is preserved under own-system rewriting], every term reachable from t stays pure. Confluence of R_1 gives w joining t_1, t_2: contradiction.

Case $n_0 > 1$. $t = C_0[s_1, \ldots, s_n]$ ($n \geq 1$). By Lemma [strict rank decrease for aliens] and Lemma [subterms of preserved terms are preserved], each s_k has rank $< n_0$ and is preserved; by minimality of n_0, each s_k is confluent for R.

Apply Lemma [reduction decomposition] to $t \to_R^* t_1$ and $t \to_R^* t_2$ (recall t is i-rooted): $C_0 \to_{R_i}^* C^{(1)}$, $t_1 = C^{(1)}[u^{(1)}]$; $C_0 \to_{R_i}^* C^{(2)}$, $t_2 = C^{(2)}[u^{(2)}]$. By Lemma [extension of confluence to extra variables], R_i is confluent with the holes as extra variables (using that R_i is confluent, whichever of R_1, R_2 it is, by hypothesis): C^* with $C^{(1)} \to_{R_i}^* C^*$, $C^{(2)} \to_{R_i}^* C^*$. By stability under substitution, $t_1 \to_R^* C^*[u^{(1)}]$, $t_2 \to_R^* C^*[u^{(2)}]$.

Each hole h of C^* traces to a unique alien origin s_k (we do not formalize here the underlying notion of tracking a subterm, possibly duplicated or erased, through a reduction: a standard but genuinely technical apparatus in rewriting theory, playing the role of Toyama's original residual-tracking proposition; we take this tracing for granted, as is common in expository accounts of this theorem); the two fillings $u_h^{(1)}, u_h^{(2)}$ are reachable from s_k, which is confluent: w_h joins them. Setting $w := C^*[(w_h)_h]$, $t_1 \to_R^* w$ and $t_2 \to_R^* w$: contradiction.

Both cases yield a contradiction, so $T = \emptyset$. ☐

Lemma (*rank is non-increasing under rewriting*). If $t \to_{R_1 \cup R_2} t'$, then $\operatorname{rank}(t') \leq \operatorname{rank}(t)$.

Proof. By well-founded induction on $\operatorname{rank}(t) \in \mathbb{N}^*$.

Base case $\operatorname{rank}(t) = 1$. t is pure (for R_1 or R_2, say R_i). By the lemma of purity is preserved under own-system rewriting, t' is pure for R_i too (the step must use a rule of R_i, since no rule of the other system can match within a term entirely over Σ_i), so $\operatorname{rank}(t') = 1 = \operatorname{rank}(t)$.

Inductive step, $\operatorname{rank}(t) = n_0 > 1$. Assume the claim for every term of rank $< n_0$. Write $t = C[s_1, \ldots, s_n]$ (i-rooted, $n \geq 1$) and apply the lemma of skeleton-alien commutation to the step $t \to t'$ at position q (dropping the preservedness proviso, since it is not assumed here):

Case (a), $p_k \leq q$ for some k. Then $s_k \to_{R_1 \cup R_2} s'_k$ and $t' = C[s_1, \ldots, s'_k, \ldots, s_n]$. Since $\mathrm{rank}(s_k) < n_0$ (lemma of strict rank decrease for aliens), the induction hypothesis applies to $s_k \to s'_k$, giving $\mathrm{rank}(s'_k) \leq \mathrm{rank}(s_k)$. If t' is pure, $\mathrm{rank}(t') = 1 \leq n_0$ trivially. Otherwise $\mathrm{rank}(t') = 1+\max\{\mathrm{rank}(s_l) : l \neq k\} \cup \{\mathrm{rank}(s'_k)\} \leq 1+\max\{\mathrm{rank}(s_l) : l \neq k\} \cup \{\mathrm{rank}(s_k)\} = \mathrm{rank}(t) = n_0$ (replacing one entry of the max by a value no larger cannot increase the max).

Case (b), preserved sub-case. $t' = C'[s_1, \ldots, s_n]$ with $C \to_{R_i} C'$, the aliens $s_1, \ldots, s_n$ unchanged as terms, only duplicated, erased, or repositioned. The multiset of alien ranks occurring in t' is thus drawn from the same values $\{\mathrm{rank}(s_l)\}$ as in t (possibly with repeats or omissions, never a new value), together with possibly new Σ_i-pure material (contributing only rank 1, never exceeding an existing alien's rank since $n \geq 1$). Hence $\mathrm{rank}(t') \leq \mathrm{rank}(t)$.

Case (b), exceptional (collapsing) sub-case. Here t' equals s_k itself, or a subterm of s_k, for some k (the alien exposed by the collapse). By the lemma of strict rank decrease for aliens, $\mathrm{rank}(s_k) < n_0$. If $t' = s_k$, we are done: $\mathrm{rank}(t') = \mathrm{rank}(s_k) < n_0 = \mathrm{rank}(t)$. If $t' \sqsubset s_k$ is a proper subterm, an entirely analogous argument to the one just given (by induction on $\sqsubset$, using that any proper subterm of a term either lies in that term's own skeleton, contributing rank 1, or is itself one of its aliens, of strictly smaller rank, by repeated application of the lemma of strict rank decrease for aliens) gives $\mathrm{rank}(t') \leq \mathrm{rank}(s_k) < n_0$. In either case, $\mathrm{rank}(t') < \mathrm{rank}(t)$.

In every case, $\mathrm{rank}(t') \leq \mathrm{rank}(t)$, completing the induction. ☐

> **Lemma** (*first-switch decomposition*). Let t be i-rooted with skeleton $t = C_0[s_1, \ldots, s_n]$, and $t \to^* t'$ any finite reduction. Exactly one holds: (N) no collapsing step exposes the currently active root, and $t' = C'[u_1, \ldots, u_k]$ with $C_0 \to^*_{R_i} C'$; (S) a type-(N) prefix $t \to^* v = C'[u_1, \ldots, u_k]$, followed by a single collapsing step exposing a filling u_{k_0}, followed by $u_{k_0} \to^* t'$, with $\mathrm{rank}(u_{k_0}) < \mathrm{rank}(t)$ and $\mathrm{rank}(t') \leq \mathrm{rank}(u_{k_0})$.

Proof. Repeated application of Lemma [skeleton-alien commutation]: as long as no step collapses the active root, every step is of type (N). The first collapsing step (if any) at the active root gives case (S). At that point, u_{k_0} is an alien of v, so $\mathrm{rank}(u_{k_0}) < \mathrm{rank}(v)$ (lemma of strict rank decrease for aliens); since $t \to^* v$ is itself a (type-(N)) reduction, chaining the lemma of rank is non-increasing under rewriting along each of its steps gives $\mathrm{rank}(v) \leq \mathrm{rank}(t)$, hence $\mathrm{rank}(u_{k_0}) < \mathrm{rank}(t)$. Similarly, chaining that same lemma along $u_{k_0} \to^* t'$ gives $\mathrm{rank}(t') \leq \mathrm{rank}(u_{k_0})$: rank never climbs back up, so no second "root-collapse relative to C_0" needs distinguishing. ☐

7.4 Proof of the modularity of confluence

Proof of the theorem on the union of confluent TRS. By well-founded induction on $\mathrm{rank}(t)$: we show, for every n_0, that every term of rank n_0 is confluent for $R := R_1 \cup R_2$, assuming (**IH**) that every term of rank $< n_0$ is confluent for R.

Case $n_0 = 1$. As in the lemma on confluence of preserved terms, Case $n_0 = 1$: t is pure for some R_i, and confluence of t reduces directly to confluence of R_i.

Case $n_0 > 1$. Let $t = C_0[s_1, \ldots, s_n]$, and consider a peak $t_1 \ {}^*\!\leftarrow t \to^* t_2$. By Lemma [first-switch decomposition], each branch is of type (N) or (S).

(N,N). Exactly the argument of the lemma on confluence of preserved terms, Case $n_0 > 1$, with (**IH**) replacing "preserved ⇒ confluent" (each alien s_k has rank $< n_0$, hence is confluent for R by **IH**, with no need for the preservation hypothesis). This gives w joining t_1, t_2.

(N,S) (the case (S,N) is symmetric). Write $t_1 = C^{(1)}[u^{(1)}]$ (type N); branch 2: $t \to^* v = C^{(2)}[u^{(2)}]$ (type N), then a collapsing step exposing $u^{(2)}_{k_0}$, then $u^{(2)}_{k_0} \to^* t_2$, with $\mathrm{rank}(u^{(2)}_{k_0}) < n_0$.

Join t_1 and v exactly as in case (N,N): $y = C^*[y_1, \dots, y_n]$ with $t_1 \to^*_R y$, $v \to^*_R y$. Since the skeleton reduction $C^{(2)} \to^*_{R_i} C^*$ never touches hole contents, y_{k_0} is the reduct obtained by joining $u^{(2)}_{k_0}$ with its counterpart on branch 1. Since $\mathrm{rank}(u^{(2)}_{k_0}) < n_0$, (**IH**) gives its confluence: applied to $t_2 \; {}^*\!\leftarrow u^{(2)}_{k_0} \to^* y_{k_0}$, this yields z with $t_2 \to^*_R z$, $y_{k_0} \to^*_R z$. We would like to conclude with the common reduct $w := C^*[y_1, \dots, z, \dots, y_n]$, giving $t_1 \to^*_R y \to^*_R w$; reaching $t_2 \to^*_R w$ from $t_2 \to^*_R z$ requires reconciling the two different scales at play here (the collapsing step has reduced t_2 down to (a descendant of) the single alien $u^{(2)}_{k_0}$, discarding the surrounding skeleton and other aliens entirely, whereas w carries that full structure alongside z); we do not spell out this reconciliation in detail, treating it, like the residual-tracing above, as a standard but genuinely technical point of the theory (it amounts to showing that the collapsing redex available at v survives, in an appropriate sense, along the skeleton reduction $C^{(2)} \to^*_{R_i} C^*$, so that y itself admits the matching collapse down to z).

(S,S). Same construction, treating both collapses symmetrically via **IH** on each side (both ranks strictly below n_0).

In every case, t_1, t_2 are joinable: t is confluent for R. By well-founded induction, every term is confluent: $R_1 \cup R_2$ is confluent. □

Nota (modularity). A property P of TRS is *modular* if, whenever R_1, R_2 both have P and $\Sigma_1 \cap \Sigma_2 = \emptyset$, $R_1 \cup R_2$ also has P. The theorem on the union of confluent TRS states precisely that confluence is modular. This modularity is far from automatic: the most instructive counterexample is termination, which is *not* modular, as we now show.

> **Proposition** (*Terminating modularity*). Termination is not a modular property.

Proof. We exhibit two disjoint terminating TRS whose union does not terminate [Toyama, 1987]. Let $\Sigma_1 = \{F, 0, 1\}$ (F ternary), $\Sigma_2 = \{G\}$ (G binary), and

$$\mathcal{R}_1 = \{\, F(0,1,x) \to F(x,x,x) \,\}, \qquad \mathcal{R}_2 = \{\, G(x,y) \to x, \quad G(x,y) \to y \,\}.$$

R_2 **terminates.** Each step replaces the redex $G(x,y)\sigma$ by $x\sigma$ or $y\sigma$, a strict subterm: $|s'| < |s|$ strictly, so no infinite reduction can exist.

R_1 **terminates.** A naive size argument fails, since $F(0,1,x) \to F(x,x,x)$ can increase size. The correct argument [Toyama, 1987] rests on the fact that $\mathcal{R}_1$ never freshly creates 0 or 1: every occurrence of the pattern $F(0,1,\cdot)$ appearing after a step was already present, in duplicated form, before it. A minimal-counterexample argument on the depth of the first redex used along a hypothetical infinite reduction then yields a contradiction; we admit this without full detail, as it is not the subject of the present article.

The union does not terminate. Let $t_0 := F(G(0,1), G(0,1), G(0,1))$. Using $G(x,y) \to x$ on the first occurrence: $t_0 \to F(0, G(0,1), G(0,1)) =: t_1$. Using $G(x,y) \to y$ on the (now second) occurrence: $t_1 \to F(0,1,G(0,1)) =: t_2$. Since t_2 matches $F(0,1,x) \to F(x,x,x)$ with $\sigma(x) = G(0,1)$: $t_2 \to F(G(0,1), G(0,1), G(0,1)) = t_0$. Hence $t_0 \to t_1 \to t_2 \to t_0 \to \cdots$ is a cyclic, infinite reduction.

Both R_1, R_2 terminate individually with $\Sigma_1 \cap \Sigma_2 = \emptyset$, yet the union admits an infinite reduction. Termination is not modular. □

Remark. The mechanism at play is exactly the one isolated by Definition [collapsing rule]: both rules of $\mathcal{R}_2$ collapse a term to one of its own arguments, allowing an alien to be repeatedly re-exposed and duplicated across the two signatures without ever terminating. This is precisely why the proof of the theorem on the union of confluent TRS tracks collapsing steps so carefully, and why it could not simply invoke Newman's Lemma: $R_1 \cup R_2$ need not terminate even when R_1, R_2 both do. The argument instead proceeds by well-founded induction on rank, a measure entirely internal to the static structure of a term, insensitive to whether the rewriting relation terminates.

Theorem (*Union of convergent left-linear TRS*). Let R_1, R_2 be convergent TRS with $\Sigma_1 \cap \Sigma_2 = \emptyset$, such that every left-hand side of every rule of $R_1 \cup R_2$ is linear. Then $R_1 \cup R_2$ is convergent.

Proof. **Confluence.** Immediate from the theorem on the union of confluent TRS, applied to R_1, R_2 (confluent, since convergent). Left-linearity plays no role in this part.

Termination. This is a substantially harder result, due to Toyama, Klop, and Barendregt (*Termination for the direct sum of left-linear complete term rewriting systems*, J. ACM 42(6), 1995). Its proof requires an apparatus considerably beyond what we have developed here: a notion of *closed* term, a well-founded order on multisets of positions, and a delicate joint analysis of collapsing and duplicating steps: precisely the combination that the proposition on non-modularity of termination shows can defeat termination in the absence of left-linearity. Left-linearity is exactly the hypothesis that rules out the kind of inconsistent, non-confluent collapsing exhibited by $\mathcal{R}_2 = \{G(x,y) \to x,\ G(x,y) \to y\}$ in that counterexample (note that this $\mathcal{R}_2$ is itself left-linear, but is *not confluent*: $G(a,b)$ has two distinct normal forms a, b; convergence, not merely left-linearity, is what is additionally required here). We do not reproduce the full argument, and instead refer the reader to the cited reference for the complete proof. $\square$

Remark. We flag this proof as *partial by design*: reproducing the Toyama–Klop–Barendregt argument in full would require substantially more space and machinery than the rest of this survey, and risks introducing subtle errors if compressed. We have chosen honesty over the appearance of completeness.

7.5 Orthogonal systems

We now turn to a second, independent route to confluence, applicable even to *non-terminating* systems, and requiring no disjointness of signatures: orthogonality. The guiding intuition is simple: if rewrite rules never overlap with one another (so a term can never be ambiguous as to which rule applies where) and never duplicate a variable on the left of a rule (so matching is unambiguous), then rewriting can be made confluent by a soft argument: any two ways of contracting a set of non-overlapping redexes in parallel can always be completed into contracting *every* redex of the term at once, and this "complete development" serves as a universal joining point.

Definition (*Orthogonal system*). A TRS R is *orthogonal* if it satisfies:

1. (*non-overlapping*) For any two rules $l_1 \to r_1, l_2 \to r_2 \in R$, there do not exist a non-variable position $p \in \mathrm{Pos}(l_2)$ and a substitution σ with $l_2|_p = l_1\sigma$.

2. (*left-linearity*) Every left-hand side is linear (no variable occurs more than once).

Lemma (*unique redex identification under orthogonality*). Let R be orthogonal and $t|_p$ a redex. Then the rule $l \to r$ and substitution σ with $t|_p = l\sigma$ are unique.

Proof. If $t|_p = l_1\sigma_1 = l_2\sigma_2$: renaming $l_2 \to r_2$ apart so that it shares no variable with l_1 (adjusting σ_2 accordingly), σ_1 and σ_2 combine into a single substitution σ (agreeing with σ_1 on $\mathcal{V}\mathrm{ar}(l_1)$ and with σ_2 on $\mathcal{V}\mathrm{ar}(l_2)$, well defined since these variable sets are now disjoint), giving $l_1\sigma = l_1\sigma_1 = l_2\sigma_2 = l_2\sigma$. Taking $p' = \varepsilon \in \mathrm{Pos}(l_1)$ (definition of overlap) exhibits this as an overlap of $l_2 \to r_2$ into $l_1 \to r_1$ at the root, forbidden unless the two rules coincide (the trivial self-overlap is not excluded by non-overlapping). Uniqueness of σ then follows from left-linearity: each variable of l occurs at a single position, so $\sigma(x) = t|_{p\cdot p_x}$ is forced. ☐

Lemma ($\Rightarrow_R^*$ *equals* $\to_R^*$). $\Rightarrow_R^* = \to_R^*$.

Proof. $\to_R \subseteq \Rightarrow_R \subseteq \to_R^*$ (Definition [Parallel rewriting]); taking reflexive-transitive closures gives $\to_R^* \subseteq \Rightarrow_R^* \subseteq (\to_R^*)^* = \to_R^*$. ☐

Definition (*complete development*). For R orthogonal, define $^\circ : \mathcal{T}(\Sigma, \mathcal{X}) \to \mathcal{T}(\Sigma, \mathcal{X})$ by well-founded recursion on $\sqsubset$:

$$t^\circ = \begin{cases} t & \text{if } t \in \mathcal{X}, \\ r\sigma^\circ & \text{if } t = l\sigma \text{ is a redex at the root, } \sigma^\circ(x) := (x\sigma)^\circ, \\ f(t_1^\circ, \ldots, t_k^\circ) & \text{if } t = f(t_1, \ldots, t_k) \text{ is not itself a redex.} \end{cases}$$

Intuitively, t° is obtained by contracting *every* redex of t simultaneously (and every redex thereby created inside the substituted variable instances, recursively).

Lemma (*strip lemma*). If $t \Rightarrow_R t'$, then $t' \Rightarrow_R^* t^\circ$.

Proof. By induction on $\sqsubset$. If t is not itself a redex and $\varepsilon \notin P$: partition P by argument and apply the induction hypothesis argument-wise. If $t = l\sigma$ is a redex and $\varepsilon \notin P$: non-overlapping forces every position of P to lie inside some variable instance $x\sigma$ (left-linearity making these instances disjoint); apply the induction hypothesis to each, then transport through the fixed skeleton l, giving $t' = l\sigma' \Rightarrow_R l\sigma^\circ$; since $l\sigma^\circ$ still matches l at the root via σ°, one further step $l\sigma^\circ \to_R r\sigma^\circ = t^\circ$ (using the same rule $l \to r$) gives $t' \Rightarrow_R^* t^\circ$ in two stages (we do not attempt to collapse this into a single parallel step, as would be needed for the sharper single-step form of this lemma found in the literature; the weaker, iterated form suffices for our purposes below, at the cost of a correspondingly more elaborate argument for confluence of $\Rightarrow_R$). If $\varepsilon \in P$: since P is an antichain, $P = \{\varepsilon\}$, and $t' = r\sigma$; joining each variable instance $x\sigma \Rightarrow_R (x\sigma)^\circ$ (trivially, via the induction hypothesis applied to the empty step) and transporting through r gives $r\sigma \Rightarrow_R^* r\sigma^\circ = t^\circ$. ☐

Lemma (*semi-diamond property for* $\Rightarrow_R$). If $t \Rightarrow_R t_1$ and $t \Rightarrow_R t_2$, there exists w with $t_1 \Rightarrow_R^* w$ and $t_2 \Rightarrow_R^* w$.

Proof. By the strip lemma applied to each step: $t_1 \Rightarrow_R^* t^\circ$, $t_2 \Rightarrow_R^* t^\circ$. Take $w := t^\circ$. ☐

Theorem (*Orthogonal confluence (Rosen)*). Every orthogonal TRS is confluent.

Proof. By $\Rightarrow_R^* = \to_R^*$, it suffices to show $\Rightarrow_R^*$ is confluent. The semi-diamond property above gives a single-step-versus-single-step peak a common reduct reachable by (possibly several) further $\Rightarrow_R$-steps on each side, rather than the sharper single-step-versus-single-step diamond available when the strip lemma itself holds in single-step form. Generalizing this semi-diamond property to arbitrary peaks $t_1 \; {}^*\!\Leftarrow_R \; t \Rightarrow_R^* t_2$ (first to a single step against an arbitrary chain, then to two arbitrary chains, each by its own induction on the length of the chains involved) is the standard next step of Rosen's argument; we do not spell out this double induction in full here, and refer the reader to Rosen's original treatment for the complete argument. Granting it, every peak of $\Rightarrow_R^*$ is joinable, so $\Rightarrow_R^* = \to_R^*$ is confluent, i.e. R is confluent. $\square$

Remark. Note the contrast with the theorem on the union of confluent TRS: orthogonality requires no disjointness of signatures and no termination assumption whatsoever, but instead requires the syntactic conditions of non-overlap and left-linearity *within a single* system. The two results address genuinely different questions and are not special cases of one another.

8 Undecidability via Turing Machines

This section is independent from the rest of the paper: it illustrates, via an encoding of Turing machines as term rewriting systems, that termination of an arbitrary TRS is undecidable. It may be skipped by a reader only interested in the confluence-theoretic development of the preceding sections.

Definition (*Turing machine*). A deterministic Turing machine is a tuple $M = (Q, \Sigma, \Gamma, \delta, q_0, q_{\mathrm{acc}}, q_{\mathrm{rej}})$, where Q is a finite set of states, Σ the finite input alphabet (with blank $\sqcup \notin \Sigma$), $\Gamma \supseteq \Sigma \cup \{\sqcup\}$ the tape alphabet, $q_0 \in Q$ the initial state, $q_{\mathrm{acc}} \neq q_{\mathrm{rej}} \in Q$ the accepting/rejecting states, and $\delta : (Q \setminus \{q_{\mathrm{acc}}, q_{\mathrm{rej}}\}) \times \Gamma \to Q \times \Gamma \times \{L, R\}$ the transition function.

Remark. We restrict attention to deterministic Turing machines throughout this work, since this is the setting needed for the reduction to term rewriting and for the undecidability results that follow. The non-deterministic case is not treated here.

8.1 Encoding the tape

We assume three distinct symbols $0, 1, \Box$ denoting a value of 0, 1, or blank. A fourth symbol $\Box_!$ denotes an infinite sequence of blanks. Denoting $c(x, l')$ a sequence starting with x and continuing with l, the tape is modeled as $\mathrm{tape}(l, s, r)$, with l, r the sequences to the left/right and $s \in \{0, 1, \Box\}$ the current symbol (we use s, not c, for the current symbol throughout this section, to avoid clashing with the list constructor c used to build tape sequences); both l, r are either $\Box_!$ or of the form $c(x, y)$. This representation is not unique; the *band normalisation* rule

$$c(\Box, \Box_!) \to \Box_!$$

removes unnecessary blanks. The moves are modeled by

$$\mathrm{left}(\mathrm{tape}(c(x, y), x', y')) \to \mathrm{tape}(y, x, c(x', y')), \qquad \mathrm{right}(\mathrm{tape}(x, y, c(x', y'))) \to \mathrm{tape}(c(x, y), x', y').$$

There are two critical pairs between each move rule and band normalisation; orienting them via Knuth–Bendix gives two additional rules, yielding a convergent system R_{tape}.

Proposition (*termination of the band normalisation rule*). $R_\square = \{c(\square, \square_!) \to \square_!\}$ is terminating.

Proof. Each step replaces a subterm of 3 function symbols by one of 1: the total symbol count strictly decreases at each step, and cannot decrease infinitely often. □

Proposition (*confluence of the band normalisation rule*). $R_\square$ is confluent.

Proof. By Newman's Lemma, it suffices to show local confluence. If two steps occur at the same position, they yield the same term. If at different positions $p \neq q$, they cannot be nested (the term $c(\square, \square_!)$ has no strict subterm equal to itself), so p, q are disjoint and the steps commute. □

8.2 Encoding configurations and transitions

Definition (*input term*). The set I_Σ of input terms is defined by: $\square_! \in I_\Sigma$; if $w \in I_\Sigma$, $a \in \Sigma$, then $c(a, w) \in I_\Sigma$.

Definition (*configuration term*). A *configuration* of M is a term $\mathrm{conf}(q, \mathrm{tape}(l, s, r))$, $q \in Q$. Given an input $w = c(a_1, \ldots, c(a_n, \square_!) \ldots)$, the *initial configuration* is $t_{M,w} = \mathrm{conf}(q_0, \mathrm{tape}(\square_!, a_1, c(a_2, \ldots)))$.

Definition (*transition rules*). For every (q, s) with $\delta(q, s) = (q', s', d)$, R_M contains (for $d = R$) $\mathrm{conf}(q, \mathrm{tape}(l, s, r)) \to \mathrm{right}(\mathrm{conf}(q', \mathrm{tape}(l, s', r)))$, or (for $d = L$) the symmetric rule with left. We further add wrapper rules pulling $\mathrm{right}(\mathrm{conf}(q, T)) \to \mathrm{conf}(q, \mathrm{right}(T))$ (and symmetrically for left), so the tape-move rules act on the tape component. R_M is the union of $R_\square$, the tape-move rules, the wrapper rules, and the transition rules.

Definition (*halting configuration*). $\mathrm{conf}(q, \mathrm{tape}(l, s, r))$ is *halting* if $q \in \{q_{\text{acc}}, q_{\text{rej}}\}$.

Lemma (*state and current symbol are unaffected by $R_\square$*). A single band normalisation step from $\mathrm{conf}(q, \mathrm{tape}(l, s, r))$ changes neither q nor s.

Proof. The rule's left-hand side is rooted by the list constructor c, distinct from conf and from any single element of Q or Γ; the redex position must therefore lie strictly inside l or r. □

Definition (*computation of M*). By induction on $k \geq 0$: $C_0 = t_{M,w}$; as long as C_k is not halting, with $\delta(q, s) = (q', s', d)$, $C_{k+1} = \mathrm{conf}(q', T^*)$, T^* the R_{tape}-normal form of the tape after the move. M *halts* on w if some C_n is halting.

Nota. This construction only uses that C_0 is a configuration term, never any property specific to the initial configuration $t_{M,w}$; we freely apply the same definition, and the propositions and lemmas built on it below, with C_0 replaced by an arbitrary configuration $C = \mathrm{conf}(q, T)$, writing "M halts from C" for the resulting notion.

Lemma (*transition subsequence*). Along a finite reduction $t_{M,w} = D_0 \to_{R_M} \cdots \to_{R_M} D_N$, at each transition step index i_j, the configuration D_{i_j+1} has the same state and current symbol as C_j, and $D_{i_j+1} \to^*_{R_{\text{tape}}} C_j$.

Proof. By induction on j, using Lemma [state and current symbol are unaffected by $R_\square$] and confluence/termination of R_{tape} to track the unique normal form of the tape through interleaved $R_\square$-steps. $\square$

Proposition (*Turing machine by a TRS*). M halts on w iff $t_{M,w}$ reduces, in R_M, to an irreducible term.

Proof. ($\Rightarrow$) Concatenate the n blocks of steps (one transition plus finitely many $R_\square$-steps) corresponding to each computation step; the final halting configuration is irreducible, since no transition rule matches an accepting/rejecting state, and its tape is already normal. ($\Leftarrow$) Given an irreducible D_N, Lemma [transition subsequence] identifies its state with that of some C_m; irreducibility forces $q_m \in \{q_{\text{acc}}, q_{\text{rej}}\}$, so C_m is halting. $\square$

Lemma (*reduction of TRS termination to TM halting*). R_M is terminating iff M halts from every configuration.

Proof. ($\Rightarrow$) Every reduction from a configuration C is finite, hence reaches an irreducible term, i.e. M halts from C (Proposition [Turing machine by a TRS]). ($\Leftarrow$) Every reduction sequence of R_M either progresses a configuration (finitely many steps, by the above) or normalises the tape (finitely many, by termination of $R_\square$). $\square$

8.3 Undecidability results

Definition (*encoding of Turing machines*). We fix an effective encoding $\langle M \rangle \in \{0,1\}^*$ of any deterministic Turing machine, identifying *algorithm* with *always-halting Turing machine*.

Theorem (*Halting problem for Turing machines*). There is no algorithm that, given an arbitrary deterministic Turing machine M, always terminates and correctly decides whether M halts on $\langle M \rangle$.

Proof. Suppose such H exists. Define D: on input $\langle M \rangle$, compute $H(\langle M \rangle)$; if "yes", loop forever; if "no", halt. Running D on $\langle D \rangle$: if D halts, $H(\langle D \rangle) =$"yes", forcing D to loop, a contradiction. If D does not halt, $H(\langle D \rangle) =$"no", forcing D to halt, a contradiction. $\square$

Theorem (*Halting problem for TRS*). There is no algorithm that, given an arbitrary TRS R, always terminates and correctly decides whether R is terminating.

Proof. Suppose such an algorithm A exists; we derive a decision procedure for "M halts on $\langle M \rangle$", contradicting the previous theorem.

A subtlety must be handled first. Termination of R_M, as we defined it, is a property of the *whole* system: by the lemma of reduction of TRS termination to TM halting, "R_M terminates" is equivalent to "M halts from *every* configuration", a substantially stronger, universally quantified statement than the specific question "M halts on $\langle M\rangle$". Reading the latter off from $A(R_M)$ only works in one direction (R_M terminating certainly implies M halts on $\langle M\rangle$, as one instance among "every configuration"); if $A(R_M)$ reports non-termination, this could stem from some unrelated configuration having nothing to do with $\langle M\rangle$, telling us nothing about the specific question we want decided.

The standard fix (Huet and Lankford [Huet and Lankford, 1978]) resolves this by refining the encoding so that non-termination of the whole system can only ever arise from one designated computation. The signature and rules are arranged so that every term other than a well-formed configuration is already irreducible (matches no rule, hence trivially terminates), and every well-formed configuration (regardless of which state or tape it encodes) is made to run M on the one fixed input $\langle M\rangle$ (rather than on its own encoded tape). This collapses the universal quantifier "every configuration" of the lemma down to the single, fixed computation of interest: the refined system, call it $R_M[\langle M\rangle]$, terminates if and only if M halts on $\langle M\rangle$. We do not reproduce this refinement in detail here; see Huet and Lankford's original construction for the complete argument. Granting it, running A on $R_M[\langle M\rangle]$ decides whether M halts on $\langle M\rangle$, contradicting the previous theorem. $\square$

Conclusion

We have surveyed the main pillars of equational reasoning via term rewriting: the notion of confluence and its two classical characterisation tools (Newman's Lemma, Section 3; the Critical Pair Theorem, Section 5, resting on the unification algorithm of Section 4); the modular behaviour of these properties under disjoint unions (Section 7), where confluence is modular but termination is not, unless left-linearity is assumed; and the principal algorithmic technique for restoring confluence, Knuth–Bendix completion (Section 6). Section 8 situates these results against the fundamental limit imposed by the undecidability of termination itself. Together, these results give a coherent picture of when, and how, rewriting can be turned into a decision procedure for equational validity, and of the genuine limits of that programme.

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